



Anatomy , physiology and surgery in neurology

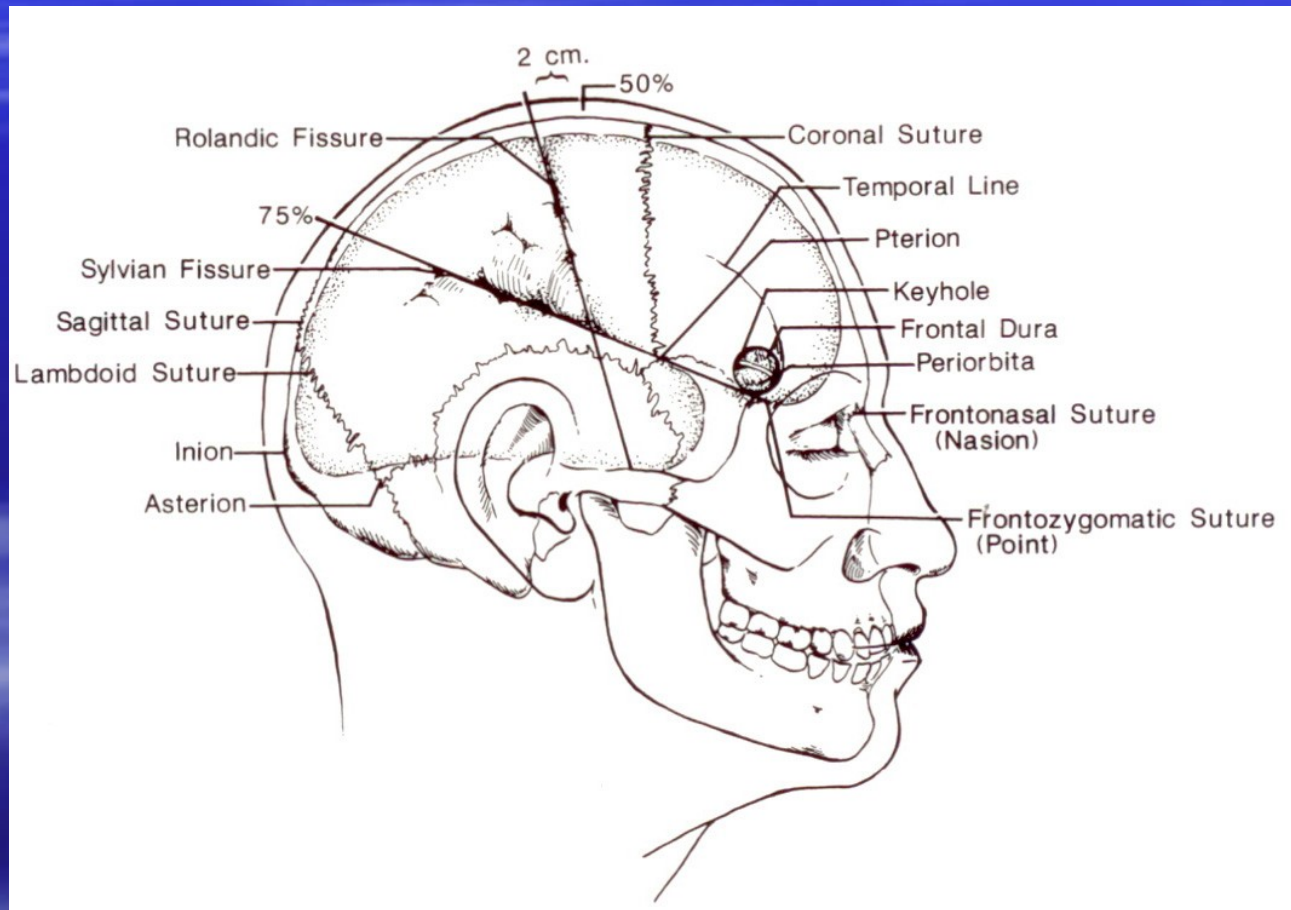
Parunut Itthimathin MD.Assist Prof.

Neurological surgery

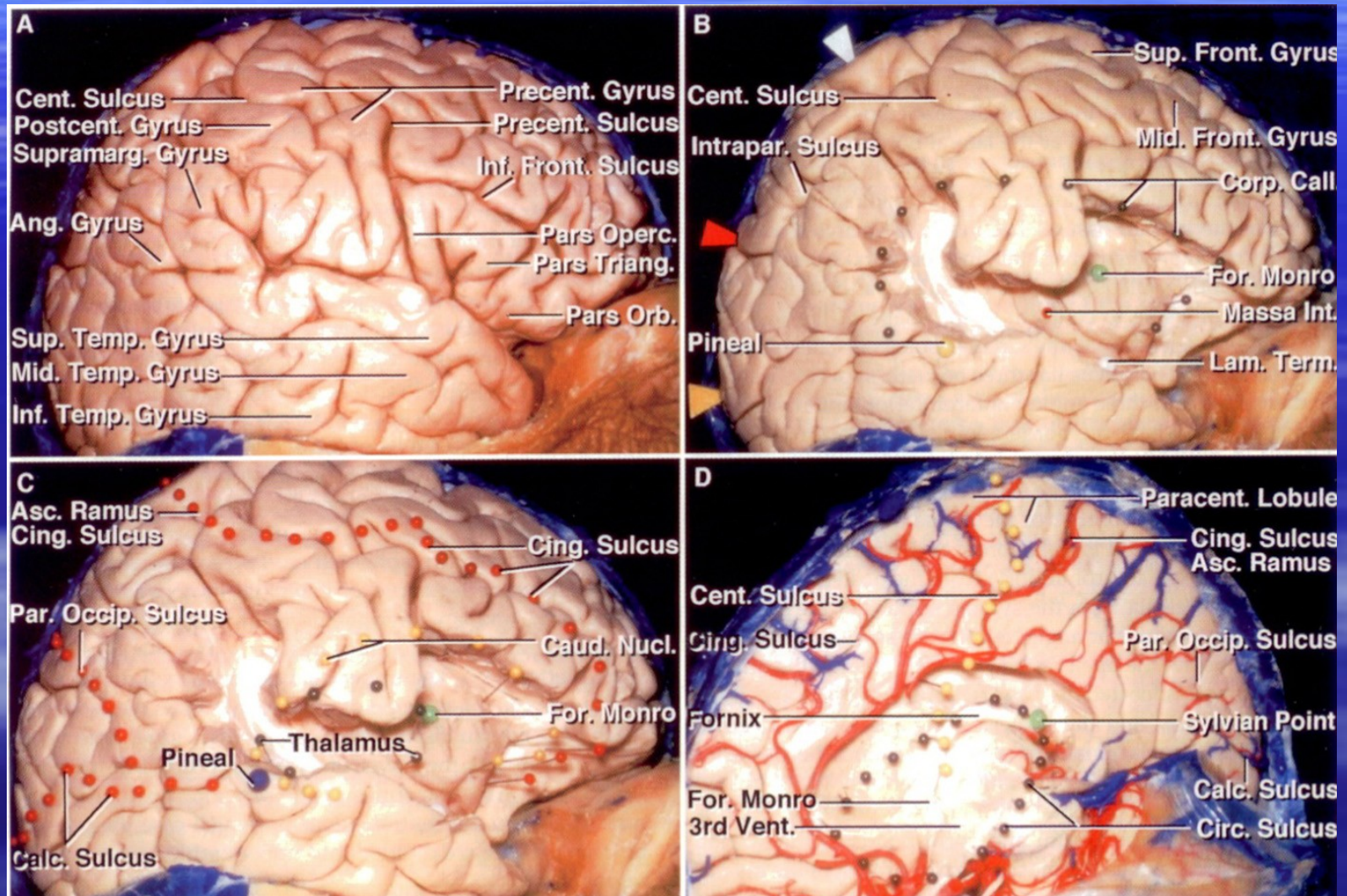
Department of surgery

Siriraj hospital

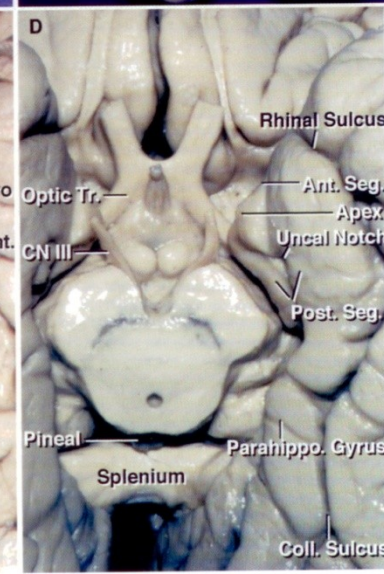
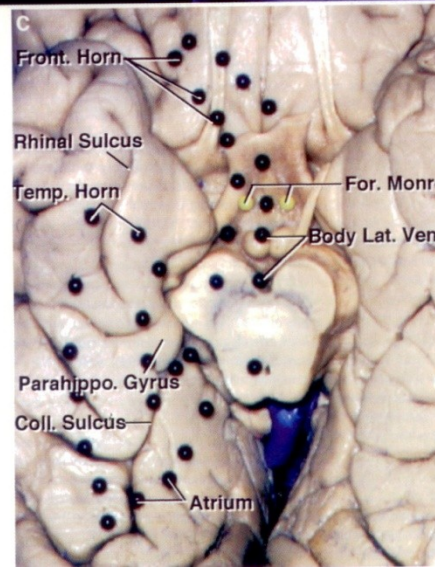
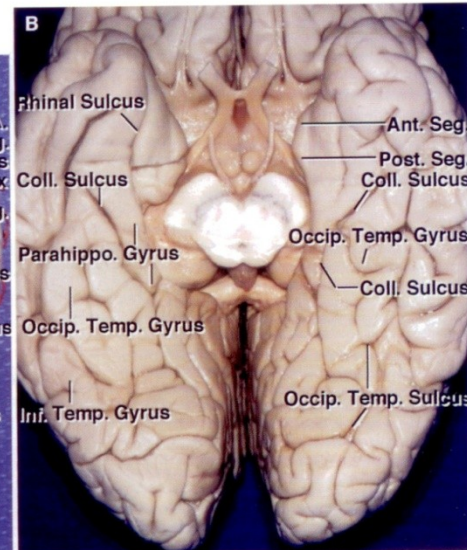
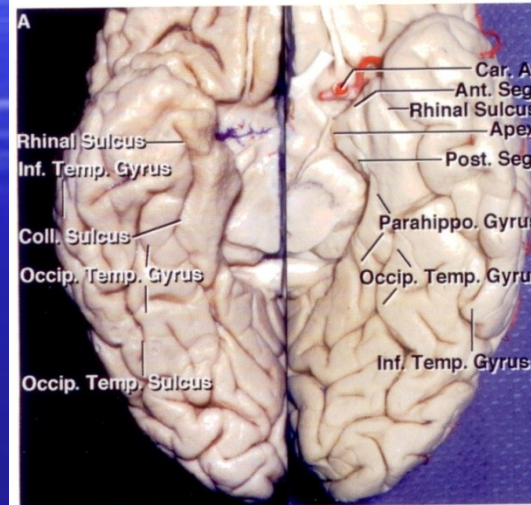
Skull anatomy



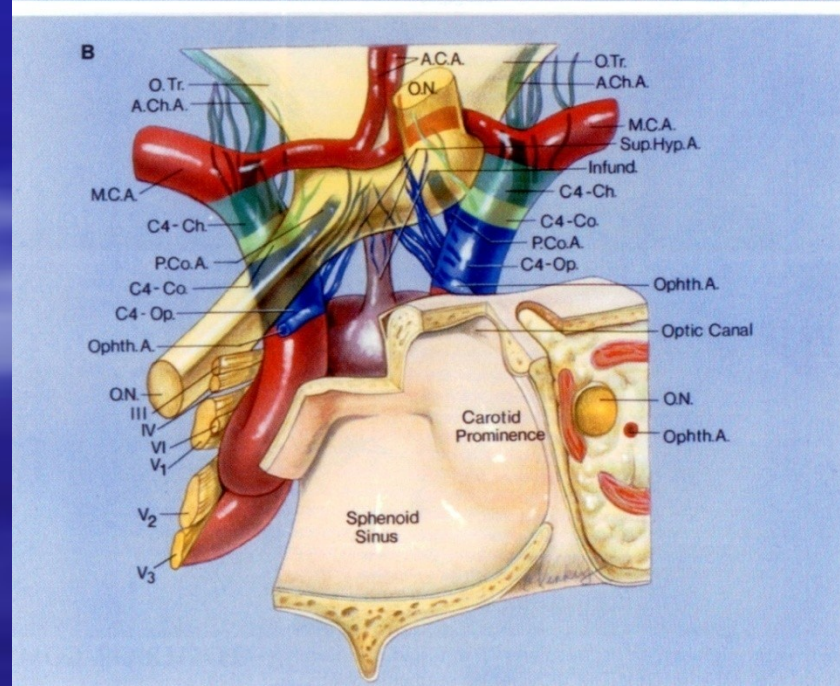
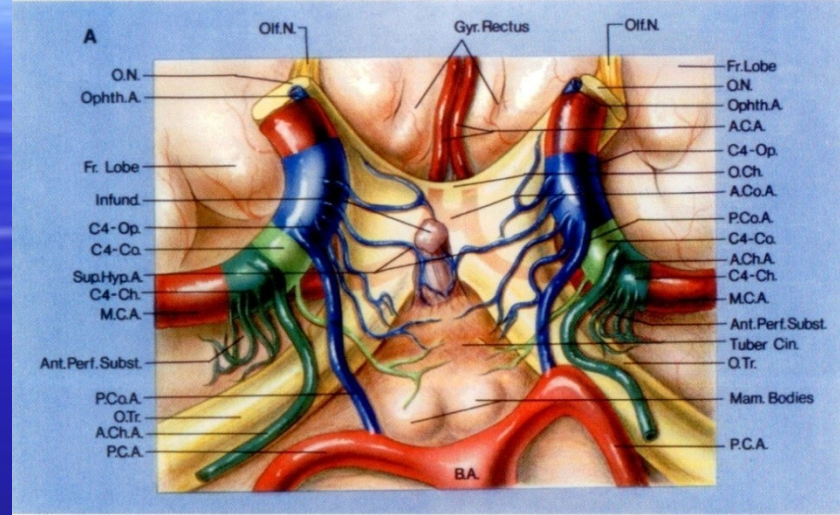
Brain surface anatomy



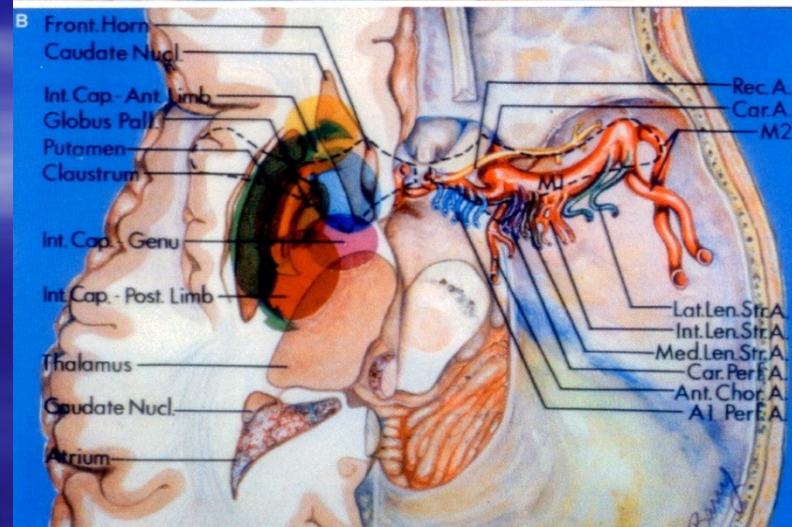
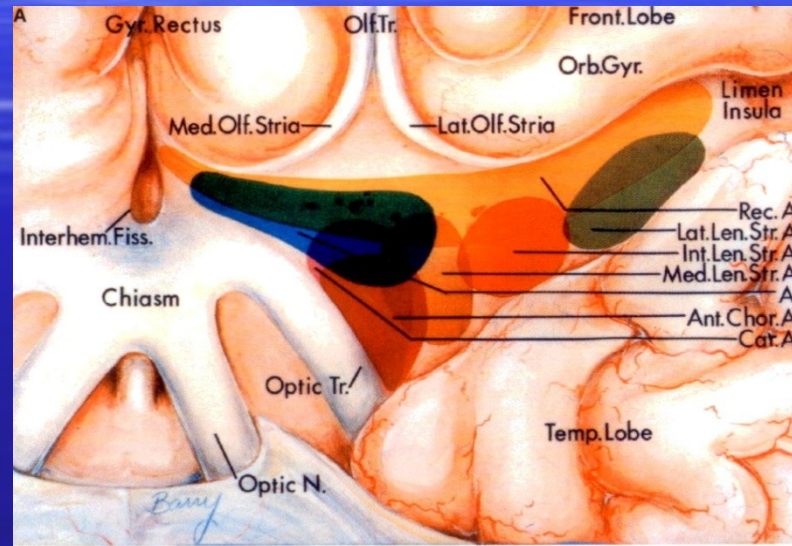
Basal surface of the brain



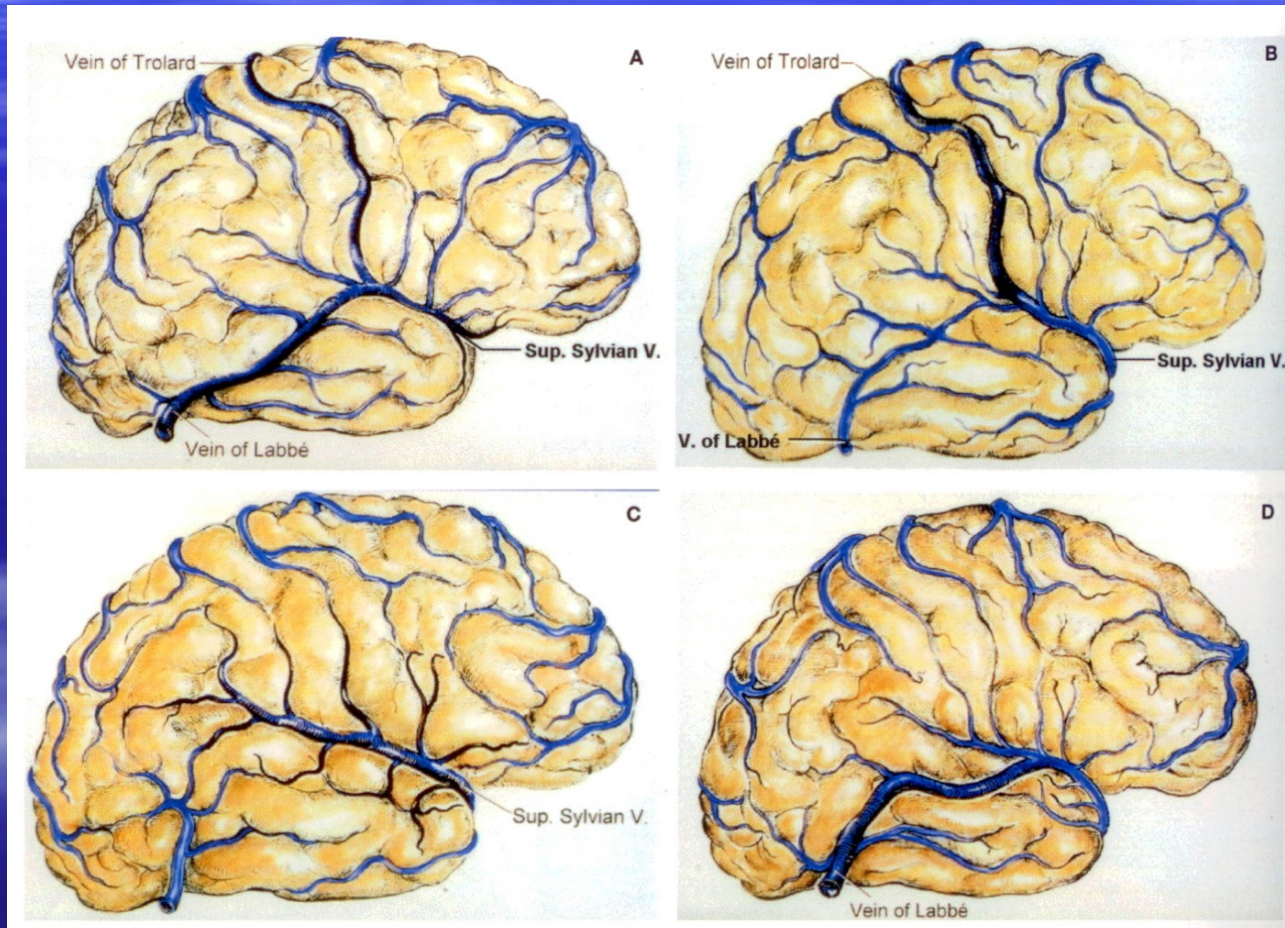
The Circle of Willis



Blood supply in deep brain

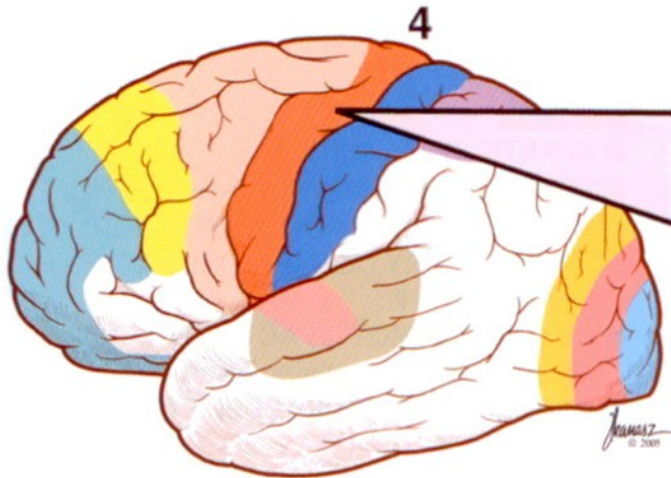


Superficial venous system of brain



Primary motor cortex

Fig. 13.8 Primary motor cortex.



LOCATION

- BRODMANN'S AREA 4

REPRESENTATION

- CONTRALATERAL HALF OF THE BODY, INVERTED

CYTOARCHITECTURE

- PREPONDERANCE OF BETZ CELLS

FUNCTION

- EXECUTION OF MOVEMENT

AFFERENT CONNECTIONS

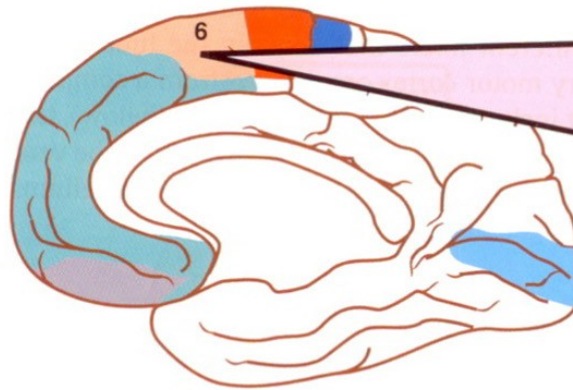
- PREMOTOR AREA
- SOMATOSENSORY CORTEX
- VL NUCLEUS OF THALAMUS

DEFICIT

- SPASTIC PARALYSIS

Supplementary motor area

Fig. 13.9 Supplementary motor area.



LOCATION

- BRODMANN'S AREA 6, MEDIAL SURFACE OF HEMISPHERE

REPRESENTATION

- PREDOMINANTLY CONTRALATERAL, FACE ROSTRAL, LEG CAUDAL

FUNCTION

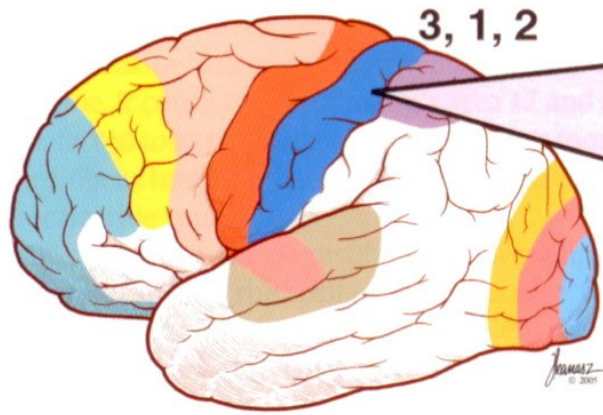
- MOVEMENT PLANNING OR PROGRAMMING

DEFICIT

- SPASTIC PARALYSIS
- TONE NORMAL OR INCREASED
- REDUCTION OF SPONTANEOUS OR VOLUNTARY CONTRALATERAL MOVEMENT
- AUTOMATIC MOTOR ACTIVITY (i.e., walking, dressing intact)
- APPEARS AS HEMINEGLECT OR APRAXIA
- MUTE
- USUALLY RESOLVES IN WEEKS TO MONTHS

Primary somatosensory cortex

Fig. 13.12 Primary somatosensory cortex.



LOCATION

- BRODMANN'S AREAS 3, 1, AND 2

REPRESENTATION

- CONTRALATERAL HALF OF THE BODY, INVERTED

CYTOARCHITECTURE

- GRANULAR CORTEX

FUNCTION

- DISCRIMINATIVE SENSATION (TOUCH, POSITION, PAIN, AND TEMPERATURE)

AFFERENT CONNECTIONS

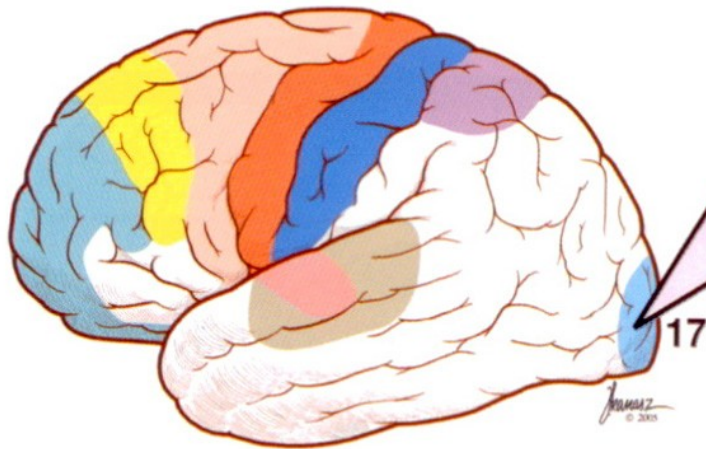
- VP NUCLEUS OF THALAMUS

DEFICIT

- IMPAIRMENT OF DISCRIMINATIVE SENSATION

Primary visual cortex

Fig. 13.15 Primary visual cortex.



LOCATION

- BRODMANN'S AREA 17

REPRESENTATION

- CONTRALATERAL VISUAL FIELD

FUNCTION

- PROCESSING OF VISUAL INFORMATION

AFFERENT CONNECTIONS

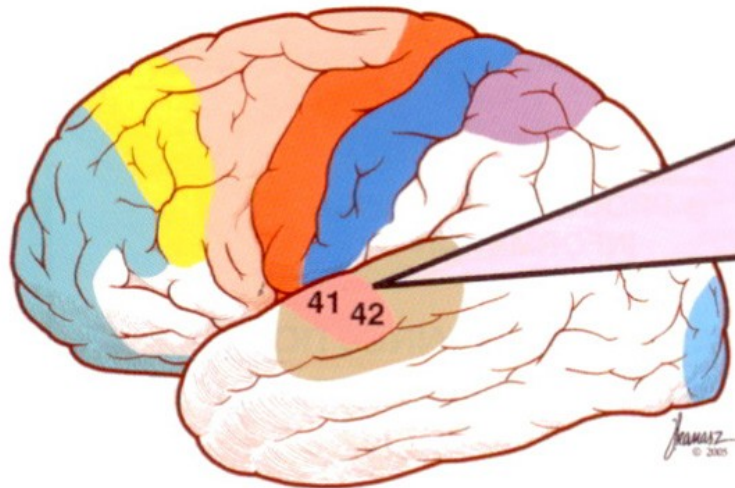
- LATERAL GENICULATE BODY

DEFICIT

- HOMONYMOUS HEMIANOPSIA WITH MACULAR SPARING

Primary auditory cortex

Fig. 13.17 Primary auditory cortex.



LOCATION

- BRODMANN'S AREAS 41 AND 42

REPRESENTATION

- BILATERAL, TONOTOPIC

CYTOARCHITECTURE

- GRANULAR CORTEX

FUNCTION

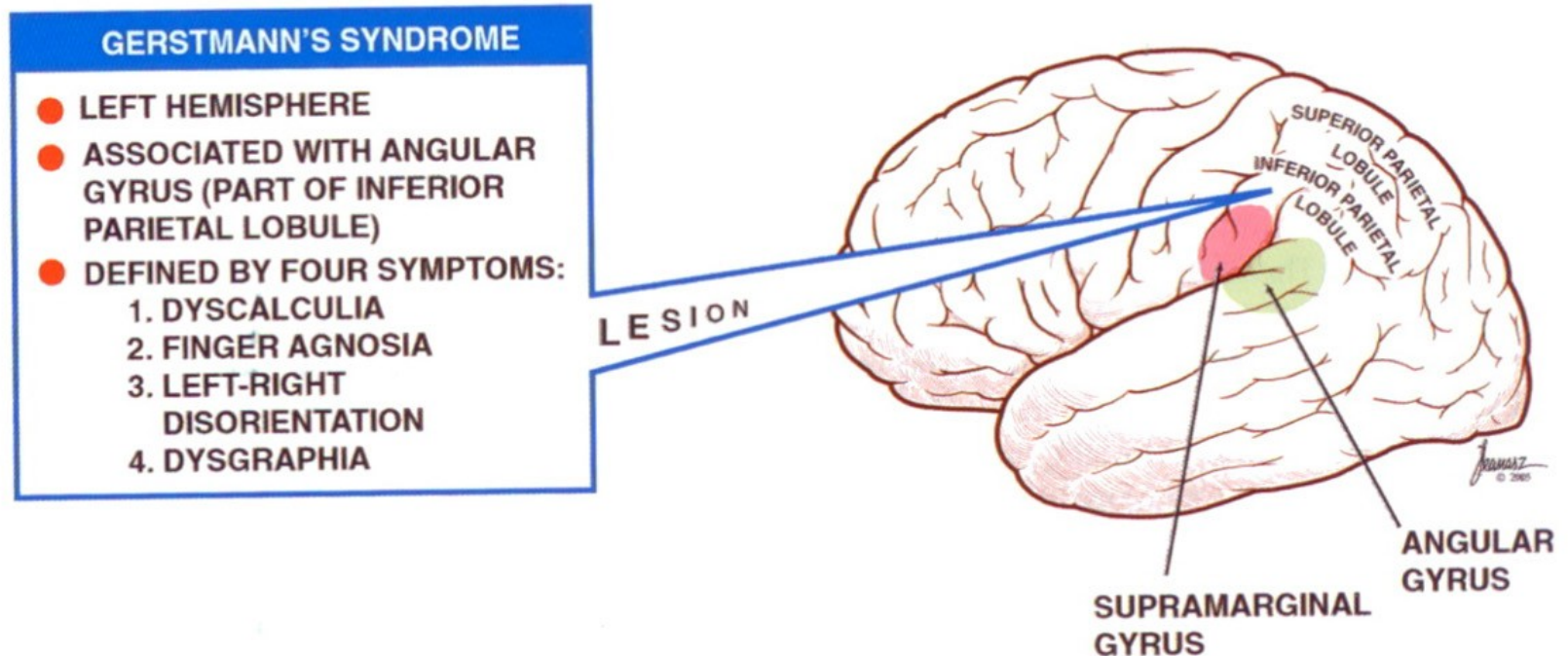
- AUDITORY PROCESSING

DEFICIT

- IMPAIRMENT IN PERCEPTION OF SOUND DIRECTION

Gerstman's syndrome

Fig. 13.25 Gerstmann's syndrome.





Localization

- Where is the lesion?
- What is the lesion?
 - congenital
 - trauma
 - tumor
 - infection
 - miscellaneous

Brain tumor

สาเหตุ ของเนื้องอกในระบบประสาท

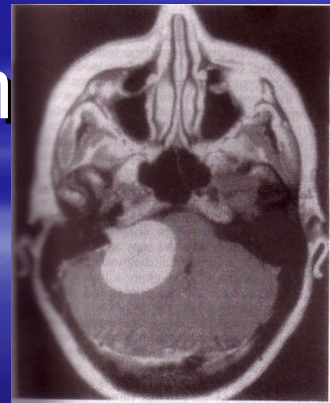
- กรรมพันธุ์
- ติดเชื้อไวรัสบางอย่าง
- สารเคมี
- ฮอร์โมนบางอย่าง
- ไม่รู้สาเหตุ

อาการและอาการแสดงของเนื้อ งอกในระบบประสาท

- ปวดศีรษะ เต้นเซ
- คลื่นไส้อาเจียน ความจำเสื่อม
- ตาพร่ามัว ถอย
- ชัก วูบ ใบหน้าเบี้ยว
- อ่อนแรง กลืนลำบาก
- พฤติกรรม เสียงแหบ
เปลี่ยนแปลง

ชนิดของเนื้องอกในระบบประสาทที่พบบ่อย

- เนื้องอกเยื่อหุ้มสมอง
- เนื้องอกต่อมใต้สมอง พิทูธิตารี
- เนื้องอกเส้นประสาท
- เนื้องอกของเนื้อสมอง
- เนื้องอกที่กระจายมาจา



Approximate distribution of central nervous system neoplasms

Histologic type	Total(%)
Glioma	50
Meningioma	20
Adenoma	15
Neuroma	8
Other	7

Data from Ng HK, et al:Aust N Z J Surg 58:573-578, 1988

Clinical feature of brain tumour

- Increased intracranial pressure
headache, visual disturbance, vomiting
- Progressive symptoms
- Seizure
- Hemiplegia or hemiparesis
- Decreased visual acuity
- Herniation syndrome
- Other

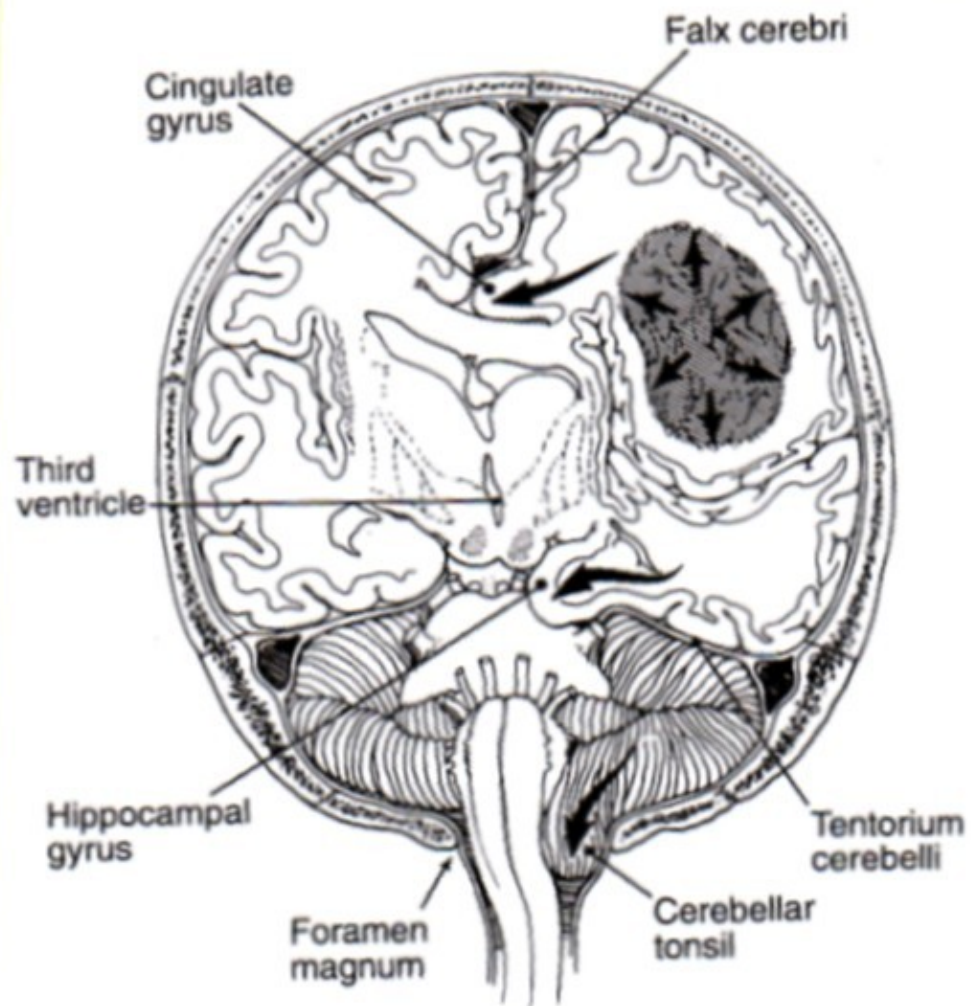


TABLE 42-1 ■ Herniation Syndromes

PATHOGENESIS	CLINICAL MANIFESTATION
Central Herniation	
Lateral and forward shift of diencephalon and upper midbrain	Decreased state of consciousness, small pupils, Cheyne-Stokes respiration
Diencephalic compression	
Midbrain pontine compression	Midposition fixed pupils Loss of oculocephalic reflexes Abnormal motor signs Hyperventilation
Lower pons—medullary compression	Decerebrate posturing leading to flaccid state Apnea Slowed pulse, raised blood pressure Death
Tonsillar (foramen magnum) herniation	
Posterior fossa mass pushes cerebellar tonsils through foramen magnum, compressing medulla	Rapid loss of consciousness, stiff neck, vomiting, skew deviation of eyes, irregular respirations, death
Uncal Herniation	
Temporal lobe herniation through tentorial notch, compressing: Ipsilateral third nerve Posterior cerebral artery Midbrain	Third nerve palsy Homonymous hemianopia Obtundation leading to coma Contralateral hemiparesis leading to decorticate and decerebrate posturing Cheyne-Stokes respiration Ipsilateral hemiparesis
Opposite cerebral peduncle	

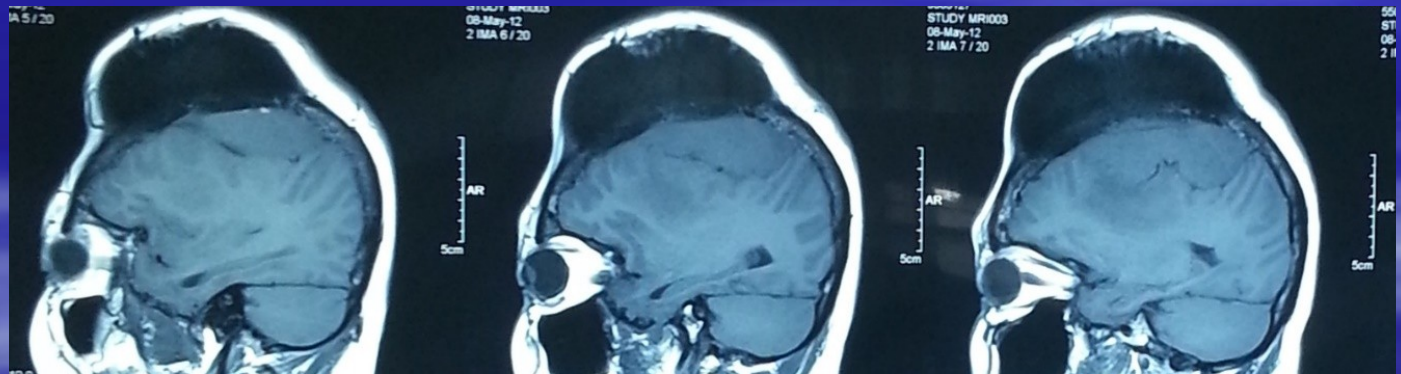
Adapted from Plum F, Posner JB: The Diagnosis of Stupor and Coma. Philadelphia, FA Davis, 1980.

**Focal Signs and Symptoms
Accompanying Brain Tumors and
Their Localization**

SYMPTOMS AND SIGNS	LOCALIZATION
Focal seizure	
Jacksonian	Centroparietal cortex
Partial (psychomotor) seizure	Temporal lobe cortex
Visual disturbances	
Altered visual acuity	Eye or optic nerve
Visual field impairment: hemianopia or quadrantanopia	Opposite lateral geniculate, optic tract, optic radiation (temporal, parietal, occipital lobe)
Bitemporal field cut	Chiasmatic (e.g., pituitary tumor)
Diplopia	Oculomotor nerves or interconnections
Nystagmus	Oculomotor nerve connections
Language and speech disturbances	Speech center(s) in dominant frontal, temporoparietal lobes
Dysphasia or aphasia (expressive and receptive)	
Dysarthria	Usually posterior fossa (cerebellum, brainstem); occasionally dominant cerebrum
Hearing disturbances	
Deafness	8th cranial nerve, cerebellopontine angle
Motor disturbances	
Weakness	Opposite corticospinal tract in cerebral hemisphere, cerebral peduncle, brainstem
Incoordination (clumsiness, dysmetria)	Posterior fossa, especially cerebellum
Sensory disturbances	
Dysesthesia	Spinal thalamic
Proprioception	Posterior spinal tracts, thalamus, thalamoparietal connections
Numbness	Thalamus
Cortical discrimination	Parietal cortex
Gait disturbances	
Weakness or sensory problems	Corticospinal tract or sensory pathways
Ataxia	Posterior fossa
Apraxia of gait	Bifrontal
Urinary disturbances	
Incontinence	Medial frontal lobes

ตรวจร่างกายและตรวจค้นเพิ่มเติม

- ตรวจร่างกายทางระบบประสาท
- เอ็กซเรย์ CT scan หรือ MRI





Brain edema

- Cytotoxic edema
- Vasogenic edema
- Interstitial edema



Treatment of brain edema

Steroids

Osmotherapy

Diuretics

Acetazolamide(Diamox)

Free radical scavengers

Selective opening of the blood brain barrier

Basic principle of cranial surgery for brain tumour

- What is the most likely pathology of the tumour?
- Does this patient need surgery?
- When should the surgery take place?
- What other radiologic or laboratory studies are necessary to be maximally prepared to carry out the surgery?

Glioma

- Low grade glioma
 - astrocytoma
 - oligodendroglioma
- High grade glioma
 - anaplastic astrocytoma
 - glioblastoma multiforme

Characteristic of Low grade glioma

- Incidence : 6-12 years and 26-46 years
- Most common presenting features is seizures
- Symptom is depend on location of tumour

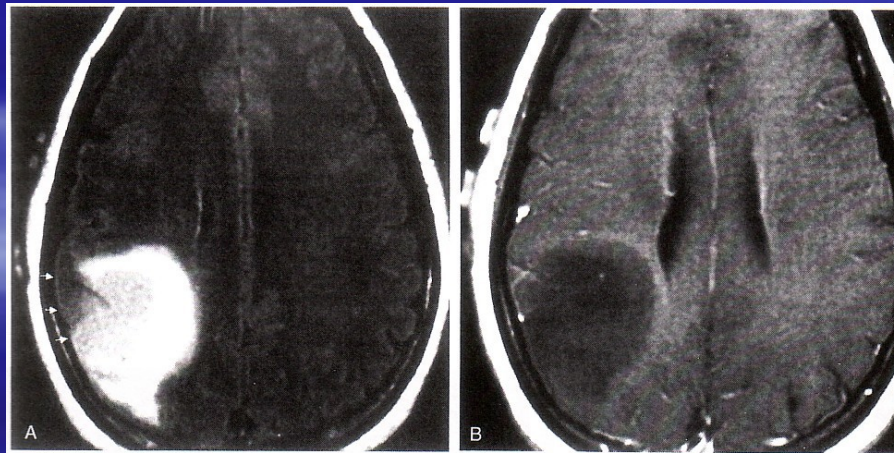


FIGURE 43-10. Low-grade astrocytoma. *A*, Notice the remodeling of bone (*arrows*) by this mass on the fluid-attenuated inversion recovery (FLAIR) scan. *B*, Absence of enhancement and the bone remodeling seen in *A* suggest that this is a low-grade lesion. Histologic analysis confirmed it was a low-grade astrocytoma.

Treatment of low grade glioma

- Surgery

 - stereotactic Biopsy

 - extensive resection

 - extensive resection with epilepsy

 - surgery

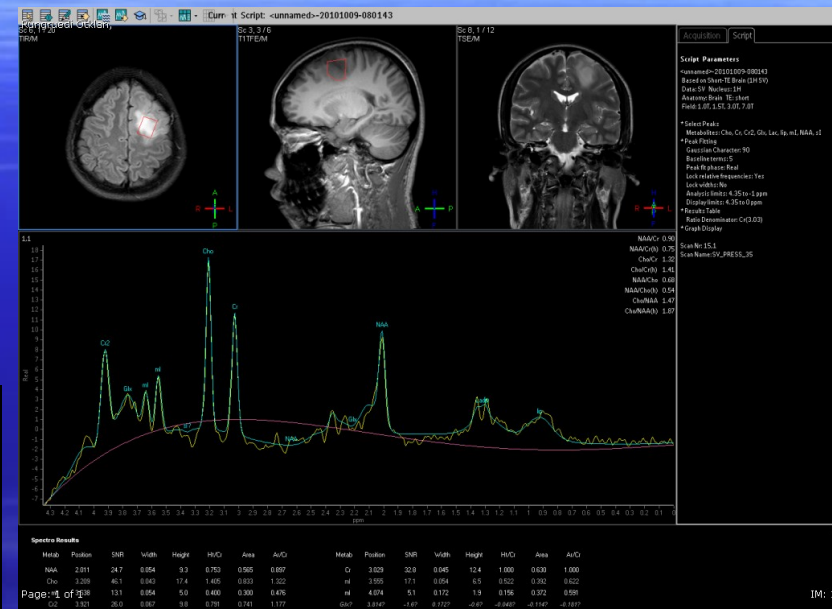
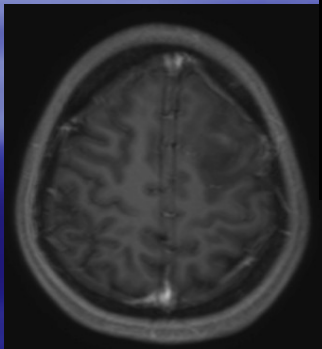
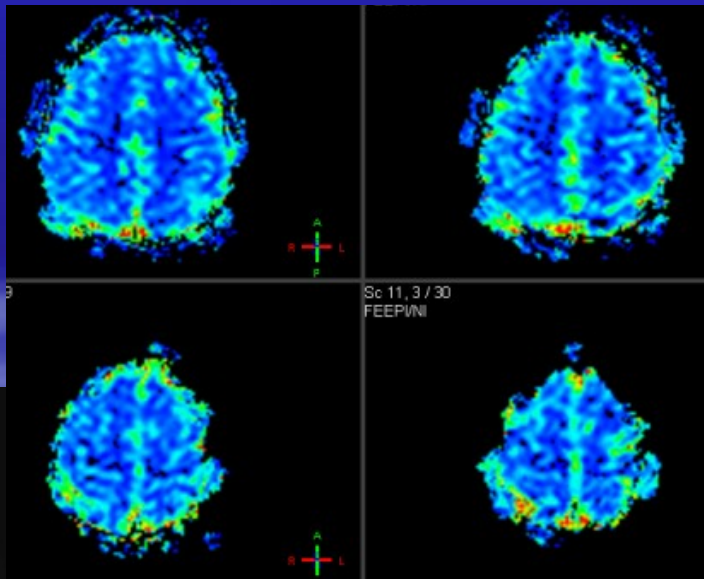
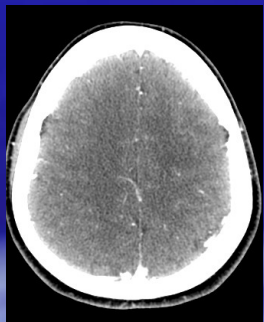
- Radiotherapy

 - controversy

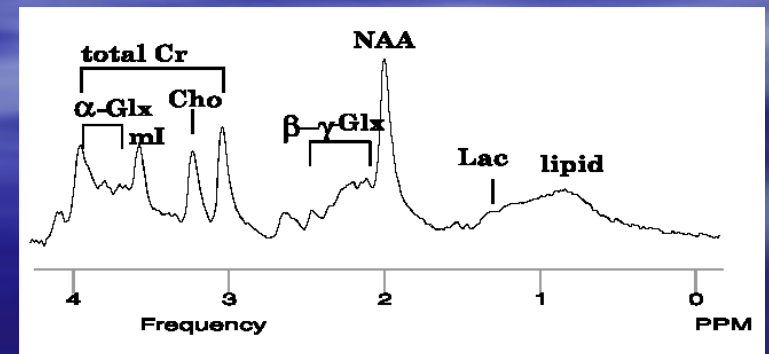
Diagnostic (pre-op) for pathological diagnosis

- Preop. neuroimaging?

✓ LGG

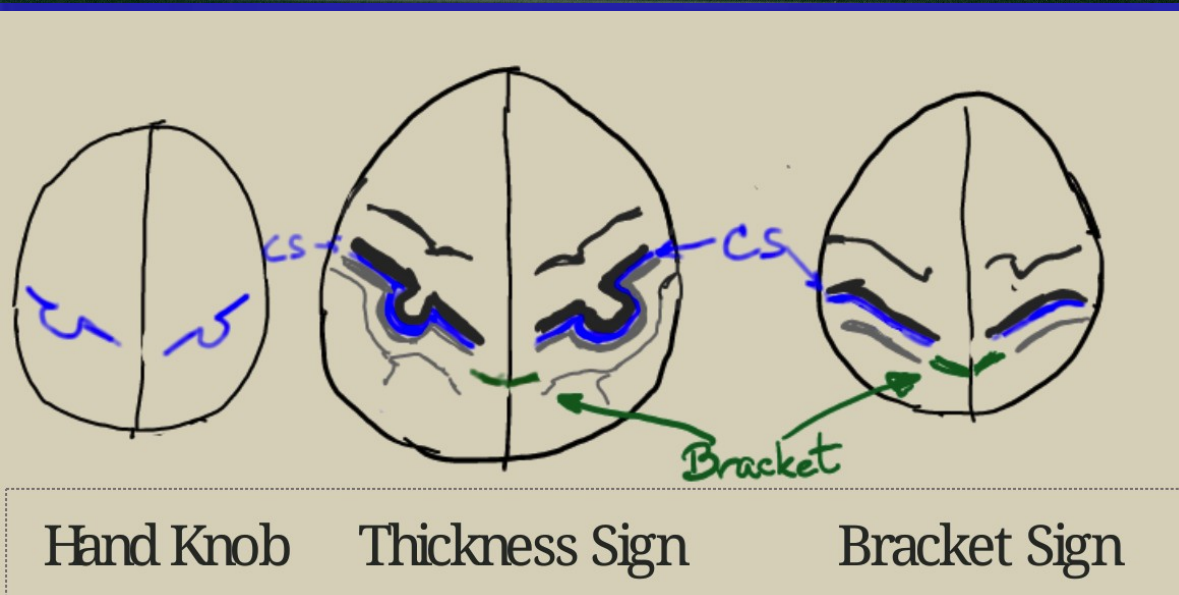
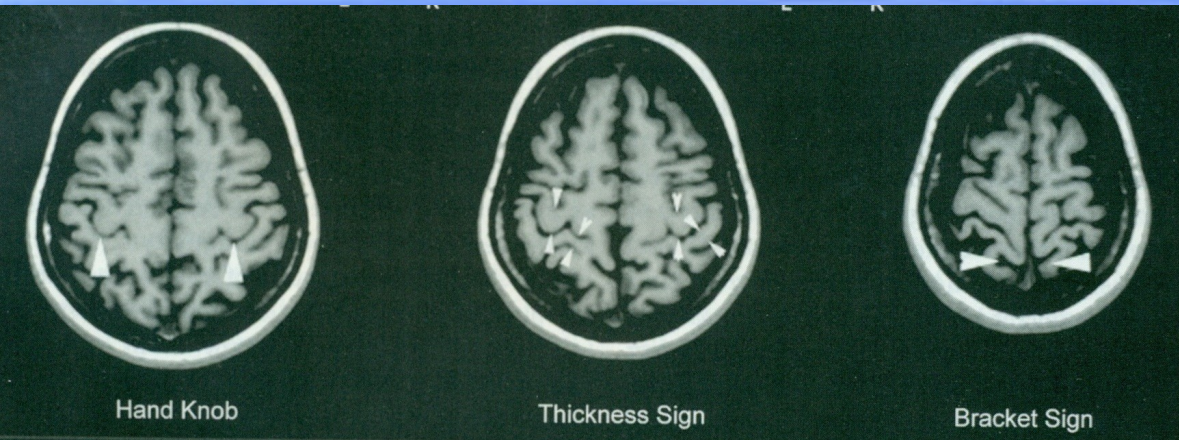


MRS: Cho:Cr 1.3-2.7



MR perfusion mild increase CBV(1.5/0.4)

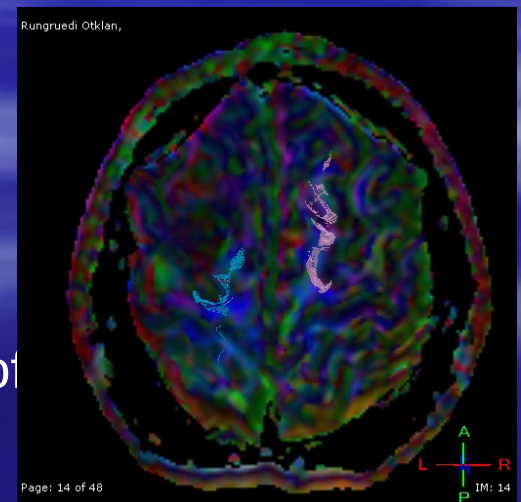
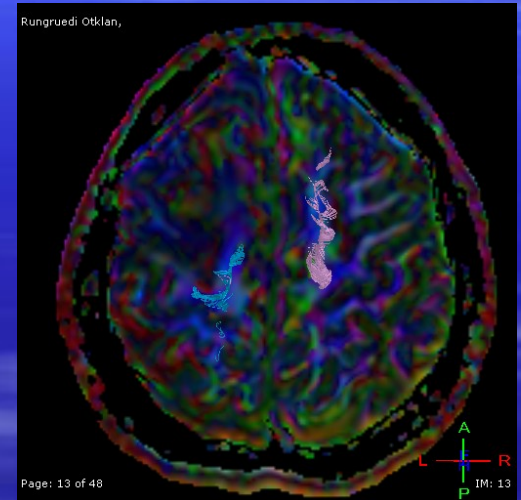
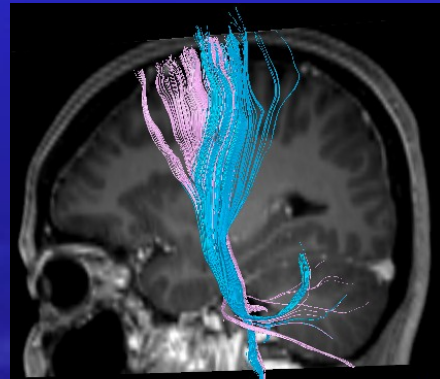
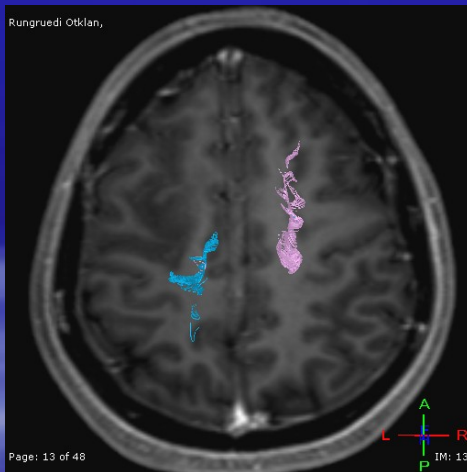
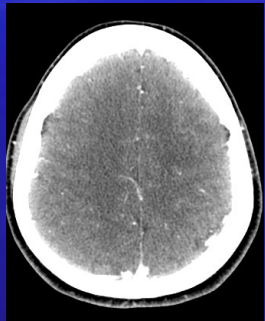
Anatomical Landmarks



Thomas P et al. Functional neuroanatomy; in Clinical functional MRI;2007: 66-68

Diagnostic (preop) for Anatomical Landmark

■ Anatomical neuroimaging



MR tactography : posterior displacement of residual left corticospinal tract

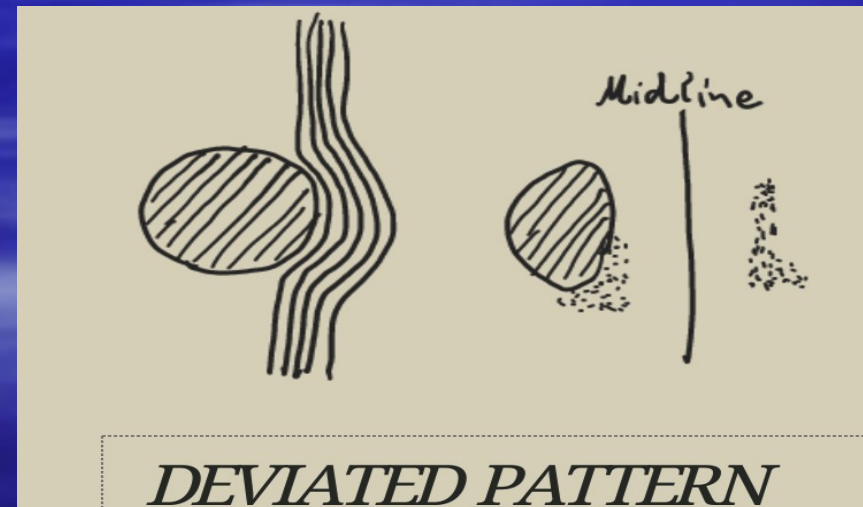
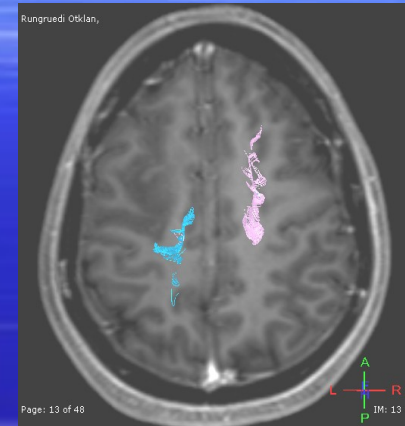
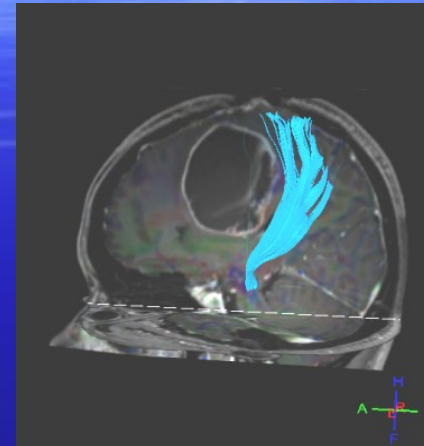
1. Deviated Pattern

Characteristic

- normal or only slightly decreased FA
- abnormal location / displacement

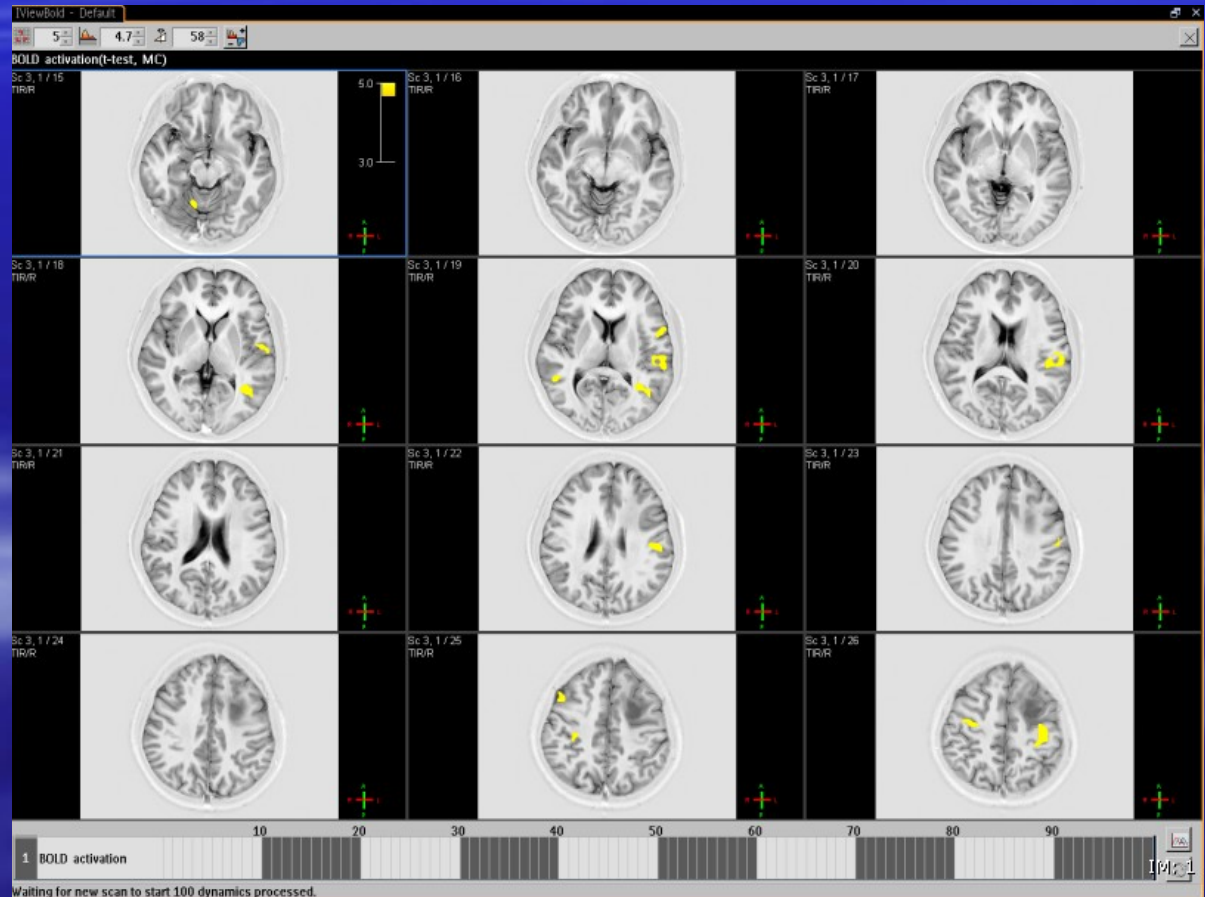
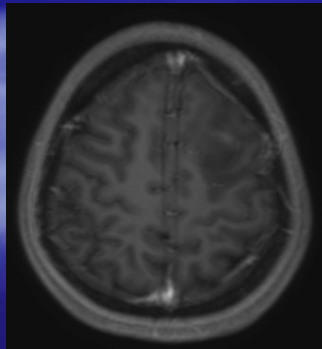
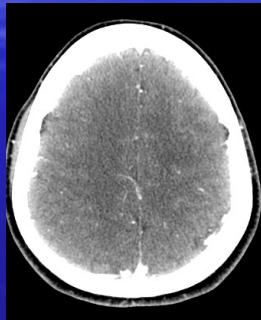
Surgical planning

- The most clinically useful pattern in preoperative planning
- Confirm intact peri-tumoral tract
- Can preserve WM tract during resection



Diagnostic (pre-op) for Anatomical Landmark

- fMRI = functional anatomy





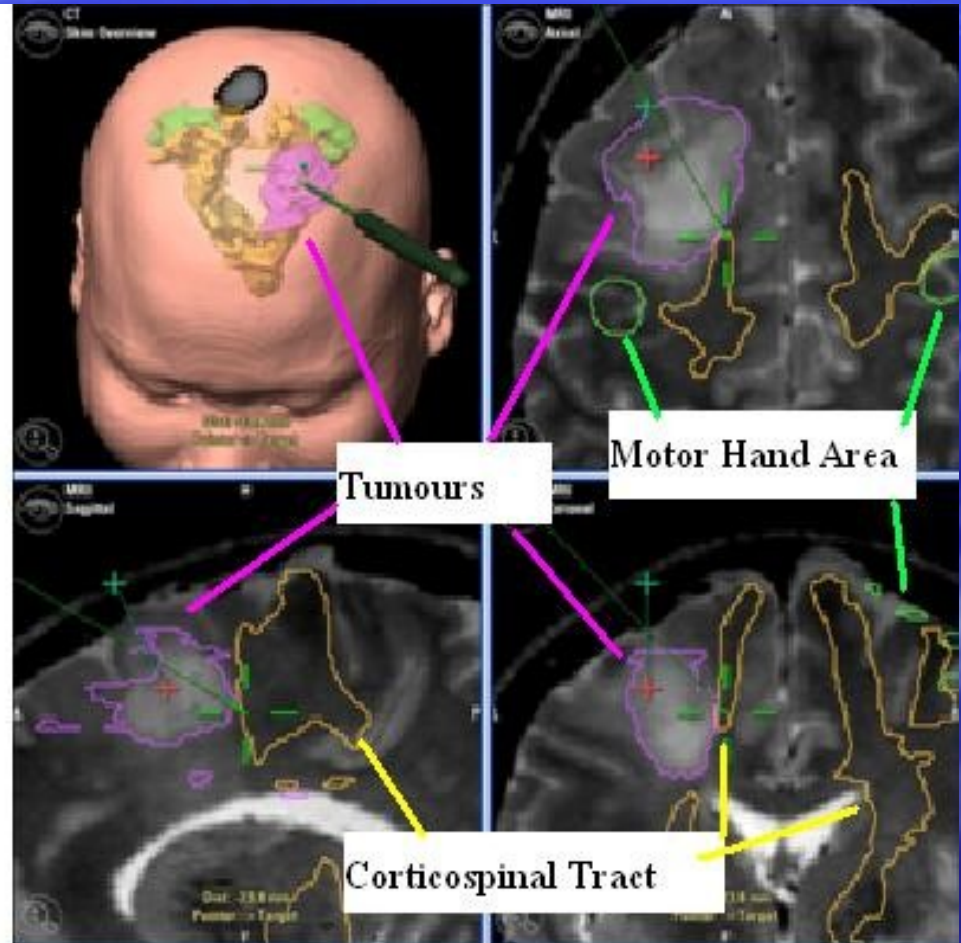
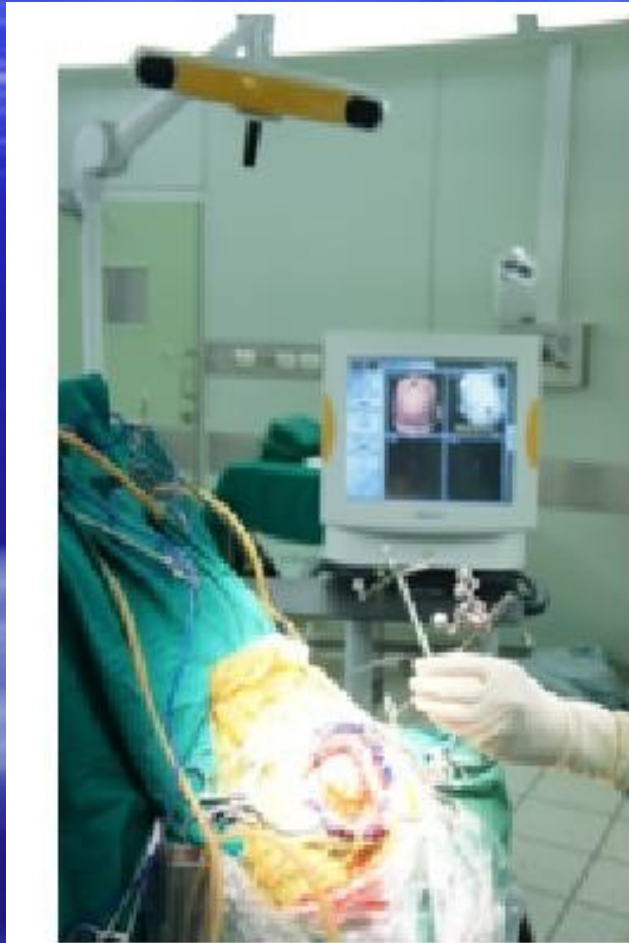
Goals of fMRI for Pre-surgical mapping

Assessment of the feasibility of surgical treatment of brain tumors

Golden rule : minimal distant for feasible resection $\geq 10\text{-}15\text{ mm}$

Distant between tumor lesion and activation	% of motor deficit
< 1 cm	50%
1-2 cm	33%
>2 cm	0%

fMRI and DTI Guided Navigation



iCT Suite: Installing in SiPH

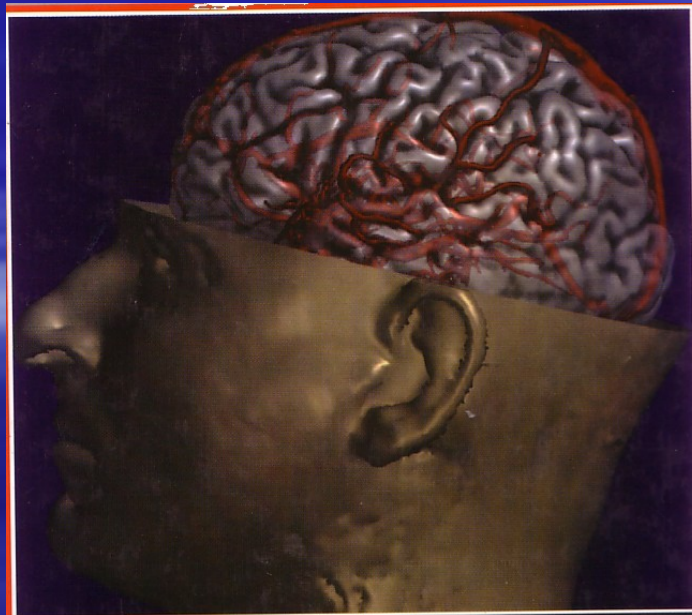


Großhadern Hospital,
Munich, Germany



Prognosis

- Median survival of 5 to 10 years
- 10 years survival rate of 5% to 50%



High grade glioma

- Glioblastoma multiforme is the most common primary brain tumour in adult.

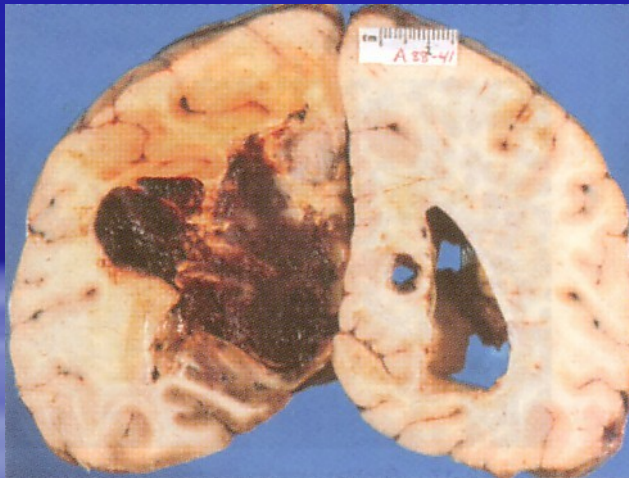


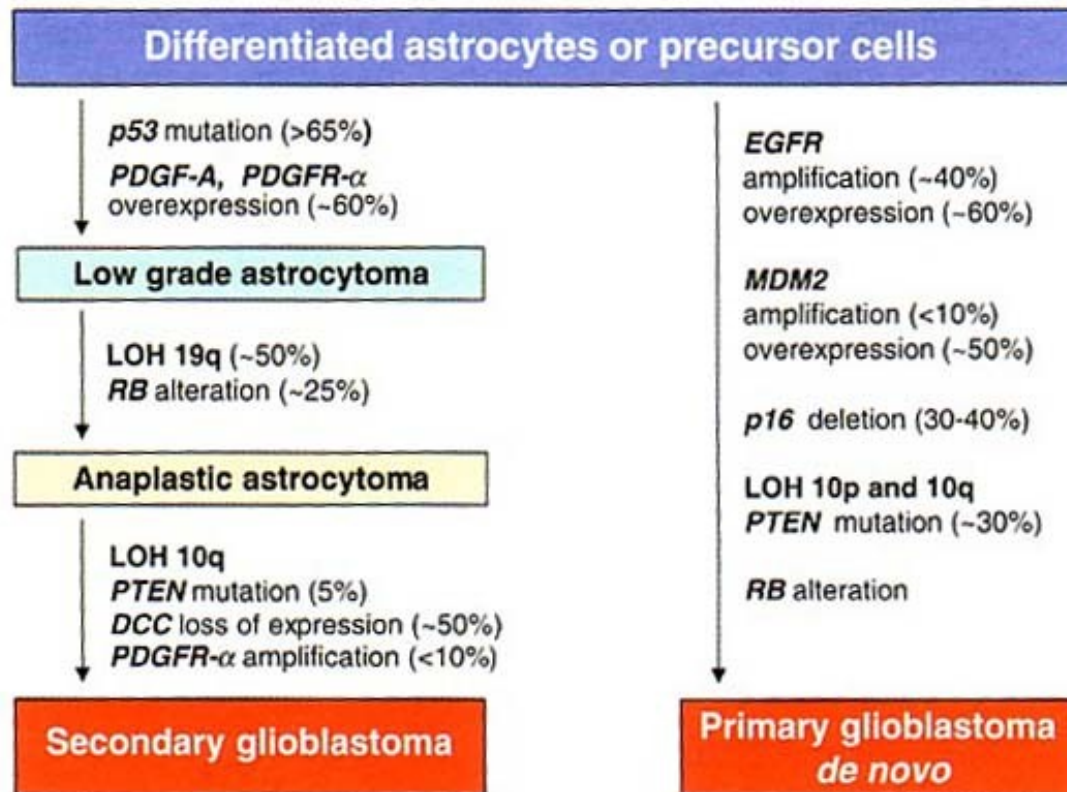
Figure 2-16. Glioblastoma multiforme. Note extensive tumor necrosis, hemorrhage, and edema.



FIGURE 4-39 GLIOBLASTOMA MULTIFORME

As evident in the axial plane of a T2-weighted image, this typical glioblastoma forms a ring-shaped mass surrounded by a broad zone of T2-bright cerebral edema (A). As seen in the frontal plane, the tumor has a necrotic center that is delineated by an irregular, shaggy rim of contrast enhancement (B). The 44-year-old patient had a 3-month history of altered mood, a 3-day history of headache, and a 1-day history of nausea and vomiting.

Molecular genetics



Clinical features of GBM

- Headache
- Focal neurological deficit
- Seizures
- Depend on tumour location

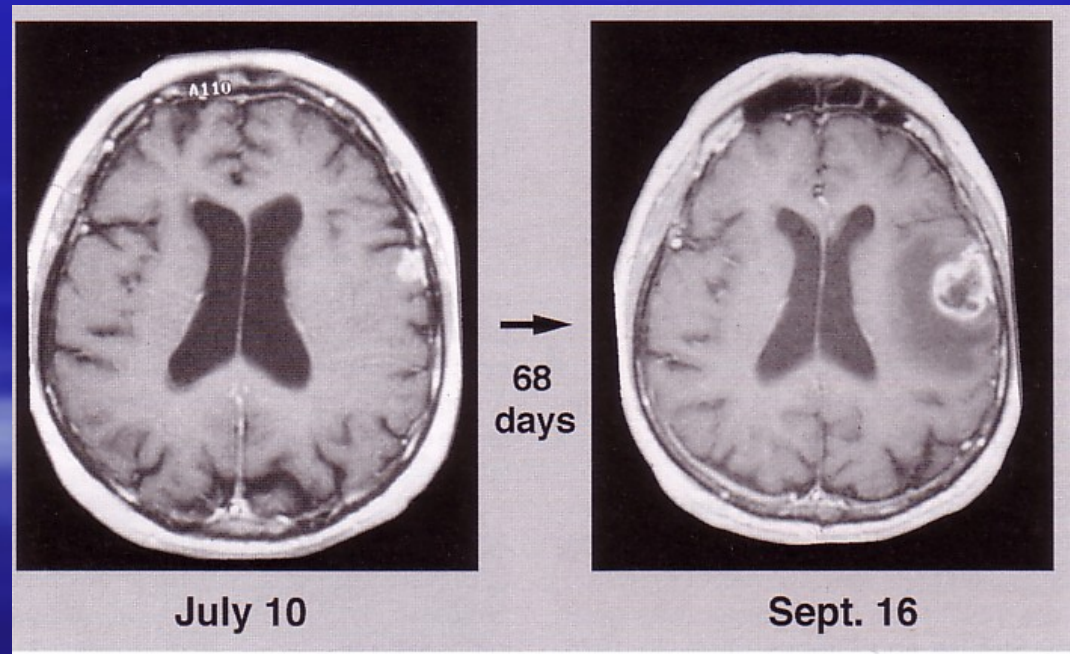


Fig. 1.23 Rapid evolution of a primary glioblastoma in a 79 year-old patient. MRI shows a small cortical lesion that within 68 days developed into a full-blown glioblastoma with perifocal oedema and central necrosis.

Diagnosis of GBM

- MRI can have more detail than CT scan and help in surgical planning.
- MRI and CT scan show ring enhanced lesion with perilesional edema
- Definite diagnosis is histopathology

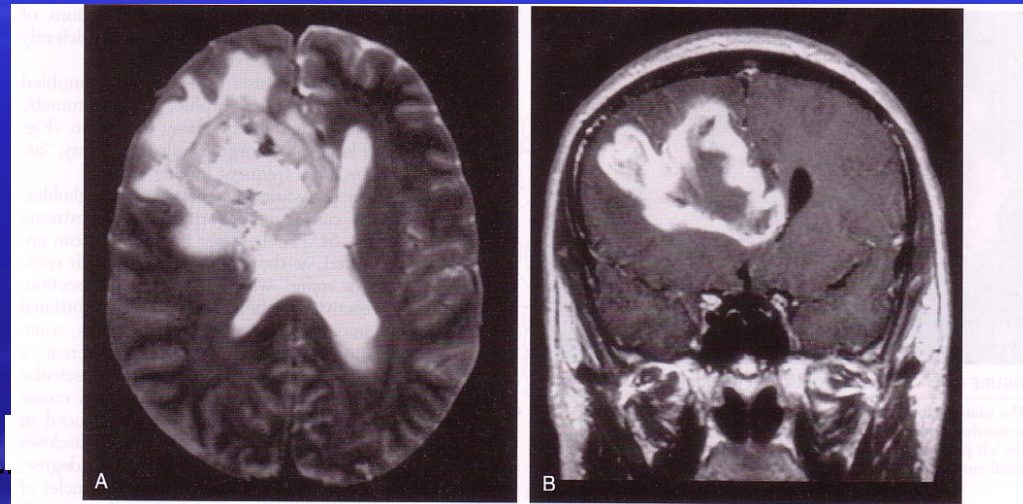


FIGURE 4-39 GLIOBLASTOMA MULTIFORME

As evident in the axial plane of a T2-weighted image, this typical glioblastoma forms a ring-shaped mass surrounded by a broad zone of T2-bright cerebral edema (A). As seen in the frontal plane, the tumor has a necrotic center that is delineated by an irregular, shaggy rim of contrast enhancement (B). The 44-year-old patient had a 3-month history of altered mood, a 3-day history of headache, and a 1-day history of nausea and vomiting.

Treatment of GBM

- Goal of surgery
 - Tissue diagnosis
 - Decrease mass effect
 - Decrease tumour burden
- Radiation therapy
 - conventional radiotherapy
- Chemotherapy
 - New drugs can improve survival of patient

prognosis

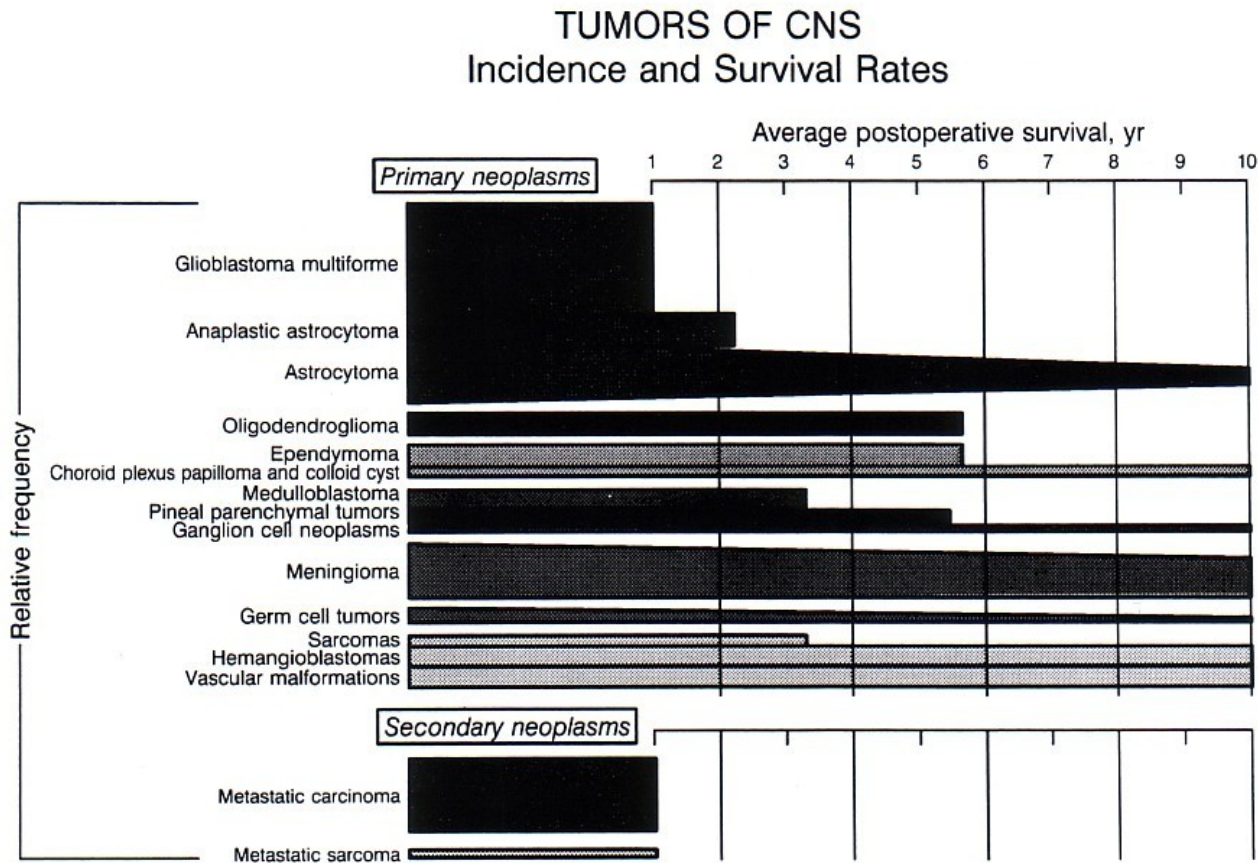
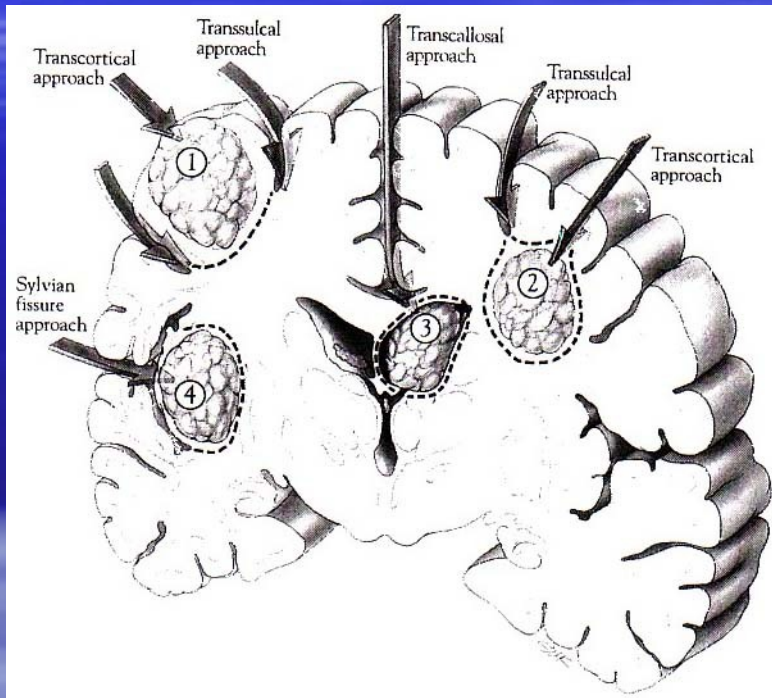


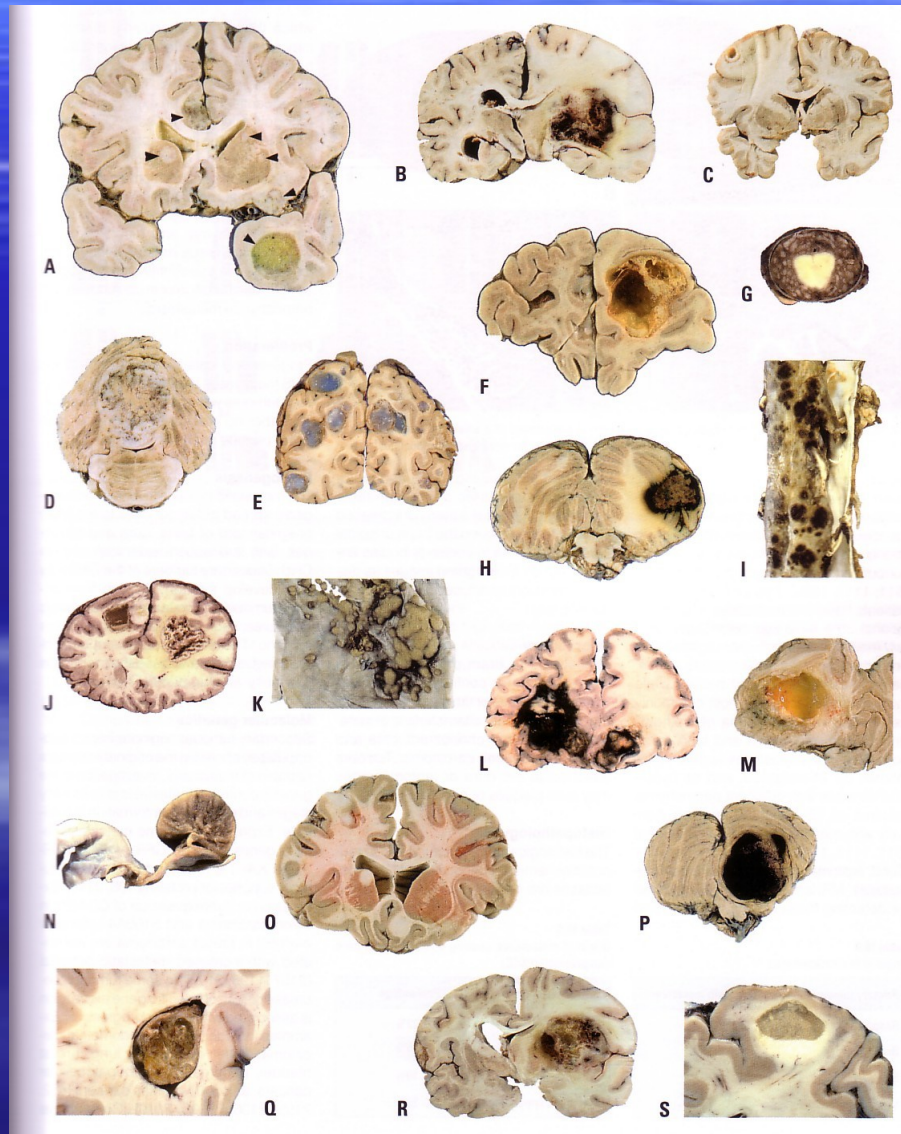
FIGURE 4-1 PRIMARY AND METASTATIC TUMORS OF THE BRAIN: INCIDENCE AND SURVIVAL RATES

Relative incidence and mean postoperative survival rates are indicated by the width and length of the individual arms, respectively.

Surgery in brain metastasis



Brain Metastasis



Origin of brain metastases

Primary tumour site	% of brain metastases
Respiratory tract	50%
Breast	15%
Skin / melanoma	10.5%
Unknown primary site	11%

Treatment of brain metastasis

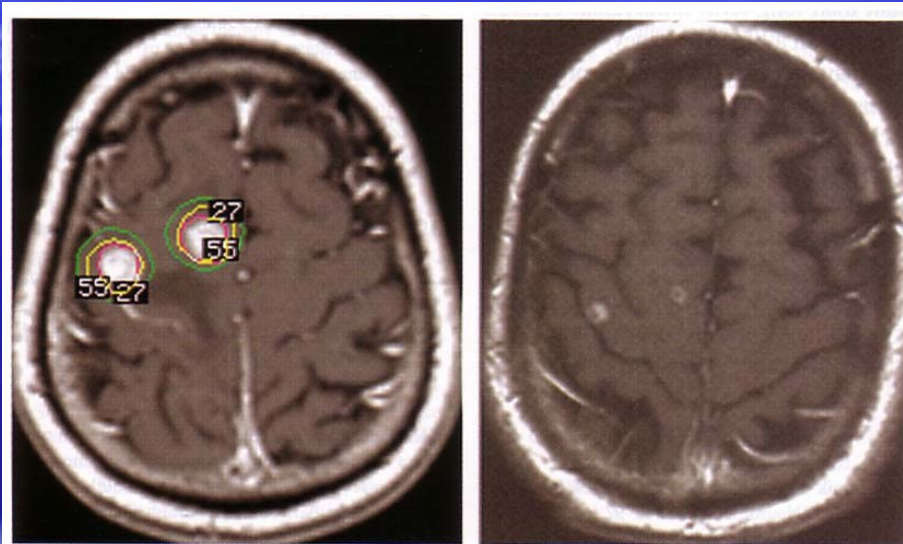


Figure 20-5. Contrast-enhanced T1-weighted magnetic resonance images on the day of radiosurgery with superimposed radiation isodose lines (*left*) and 6 months after radiosurgery (*right*), showing partial response of two metastases treated with 18.0 Gy at the 55% isodose line.

- Surgery + Whole brain radiation
- Whole brain radiation alone if it has multiple lesions or contraindication in surgery
- Radiosurgery

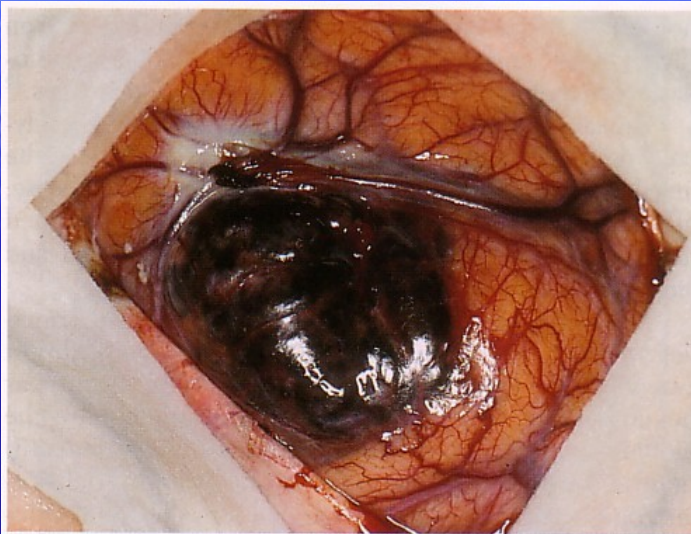


FIGURE 4-508 METASTATIC MELANOMA

Metastases are discrete and often superficial. This black lesion had its origin in the choroid of the eye. (Courtesy of Dr. Henry Brem, Baltimore, MD.)



FIGURE 4-509 METASTATIC CARCINOMA

Metastases are often fleshy, largely necrotic, and discrete.

TABLE 61-5 ■ Resection and Whole-Brain Radiation Therapy versus Radiosurgery and Whole-Brain Radiation Therapy

AUTHOR AND YEAR	NO. OF PATIENTS	MEDIAN SURVIVAL (WK)	CNS DEATH RATE (%)	LOCAL CONTROL (%)
S+ WBRT				
Patchell et al 1990 ⁸⁰	25	40	29	80
Noordijk et al 1994 ¹²⁶	32	43	35	NA
Mintz et al 1996 ⁸⁶	41	22.4	NA	NA
RS+ WBRT				
Auchter et al 1996 ¹²⁵	122	56	25	87
Shu et al 1996 ¹⁷⁸	23	70	NA	NA

CNS death, death due to central nervous system causes; NA, not available; RS, stereotactic radiosurgery; S, surgery; WBRT, whole-brain radiation therapy.

From Boyd TS, Mehta MP: Stereotactic radiosurgery for brain metastases. *Oncology* 13:1397-1410, 1999.



■ Meningioma

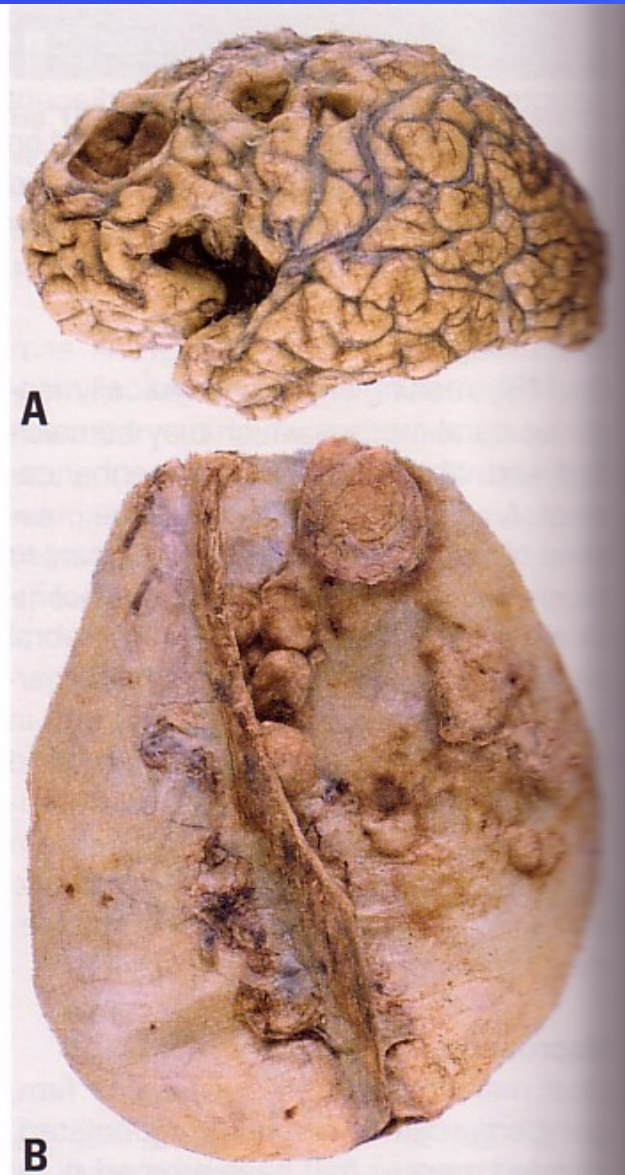


Fig. 11.6 Multiple meningiomas (**B**) of the hemispheric convexity and (**A**) the impressions caused in the left cerebral hemisphere.

Meningioma

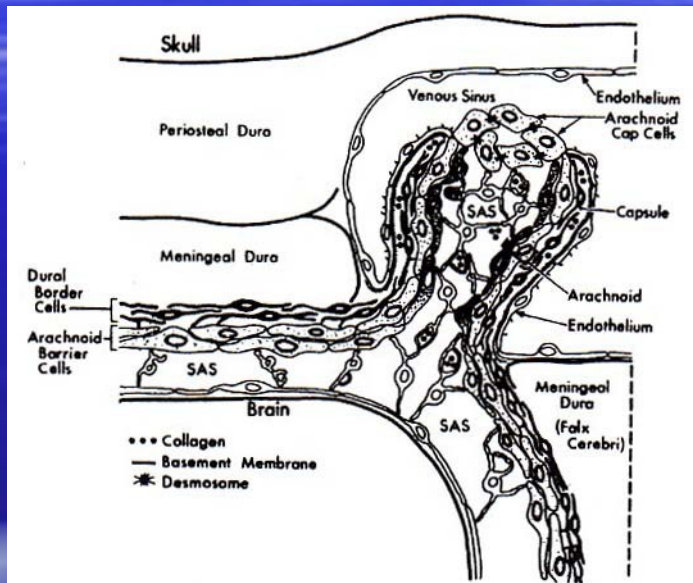
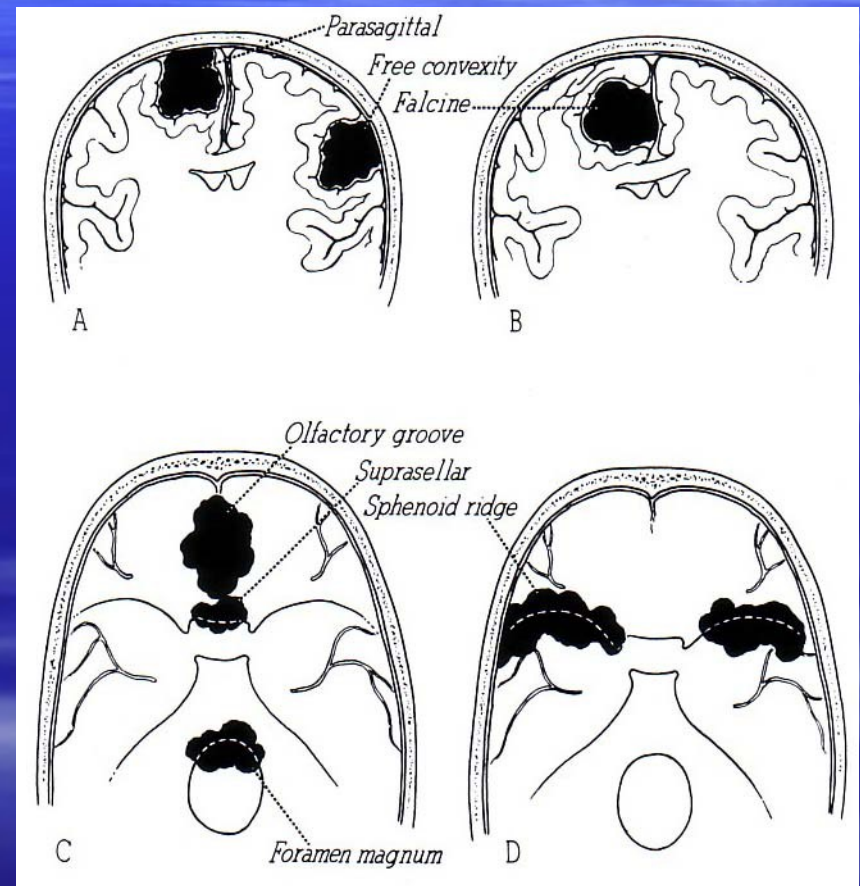


FIGURE 62-1. Semidiagrammatic representation of the arachnoid granulation shows its relation to the superior sagittal sinus (venous sinus) and the continuity of its layers and spaces with those of the meninges on the surface of the brain. The individual layers or cell clusters are not intended to be proportional but are drawn to emphasize the cytoarchitectural features of the granulation. SAS, subarachnoid space. (From Haines DE, Frederickson RG: The meninges. In Al-Mefty O (ed): Meningiomas. New York, Raven Press, 1991, p 9.)



arachnoidal cells

NF2 gene mutation/chromosome 22q loss
? other loci

meningioma, WHO grade I

losses of 1p, 6q, 10q, 14q and 18q
gains of 1q, 9q, 12q, 15q, 17q and 20q,

atypical meningioma, WHO grade II

losses of 6q, 9p, 10 and 14q
17q amplification
rare mutations: TP53, PTEN
rare deletions: CDKN2A

anaplastic (malignant) meningioma, WHO grade III

Fig. 11.12 Genetic changes associated with meningioma progression.

Classification of meningioma

TABLE 62-2 ■ Meningiomas Grouped by Likelihood of Recurrence and Grade

Grade I

Meningothelial meningioma
Fibrous (fibroblastic) meningioma
Transitional (mixed) meningioma
Psammomatous meningioma
Angiomatous meningioma
Microcystic meningioma
Secretory meningioma
Lymphoplasmacyte-rich meningioma
Metaplastic meningioma

Grade II

Atypical meningioma
Clear cell meningioma (intracranial)
Chordoid meningioma

Grade III

Rhabdoid meningioma
Papillary meningioma
Anaplastic (malignant) meningioma

Adapted from Louis DN, Scheithauer BW, Budka H, et al: Meningiomas. In Kleihues P, Cavenee WK (eds): *Pathology and Genetics of Tumours of the Nervous System*. Lyon, IARC Press, 2000, pp 176-180.

Etiology of meningioma

- Trauma
- Viruses: inoue-Melnick virus(IMV)
- Irradiation
- Other associations :
 - breast cancer and meningioma
 - Venous thrombosis of legs and meningioma

Clinical presentation of meningioma

- Depend on location of tumour
- Slow progression
- Symptom of increase intracranial pressure
- Neurological deficit
- Seizure
- other

Table 18–1. COMMON LOCATIONS FOR MENINGIOMAS

Clinical Series	No. of Patients	Location (% of Meningiomas Occurring)			
		Convexity	Parasagittal	Sphenoid Wing	Posterior Fossa
Cushing and Eisenhardt ¹	295	18	22	18	8
Jaaskelainen ²¹	657	25	21	12	3
Kallio et al ²²	935	22	27	23	10
Stafford et al ²³	581	16.8	18.6	15.3	13.6

Table 18–2. GROUPING OF MENINGIOMAS BY PROBABLE RECURRENCE AND GRADE*

Histopathologic Subtype	Risk of Recurrence / Aggressive Growth†	WHO Grade
Meningothelial	Low	I
Fibrous	Low	I
Transitional	Low	I
Psammomatous	Low	I
Angiomatous	Low	I
Microcystic	Low	I
Secretory	Low	I
Lymphoplasmacyte-rich	Low	I
Metaplastic	Low	I
Atypical	Moderate	II
Clear cell	Moderate	II
Chordoid	Moderate	II
Rhabdoid	High	III
Papillary	High	III
Anaplastic	High	III

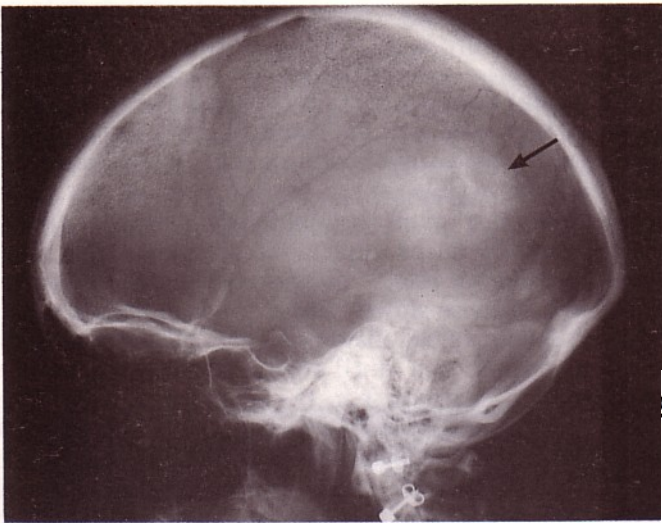
*World Health Organization grading system, 2000.

†Higher for any subtype with high proliferation index and/or brain invasion.

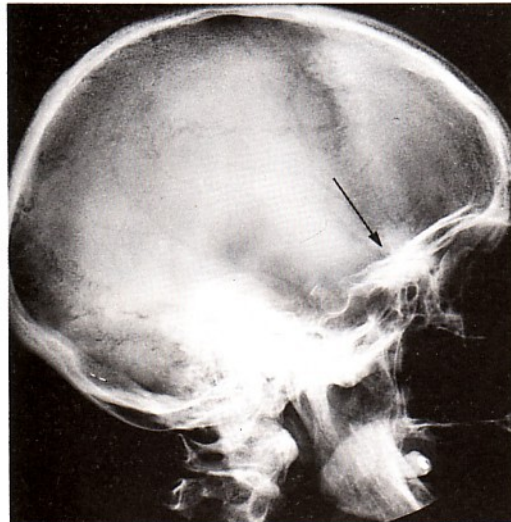
Adapted from Kleihues P, Cavenee WK, International Agency for Research on Cancer, editors. Pathology and genetics of tumours of the nervous system. Lyon: IARC Press; 2000.

X-ray in meningioma

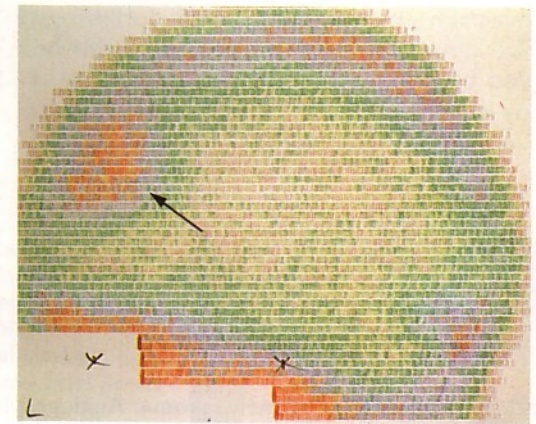
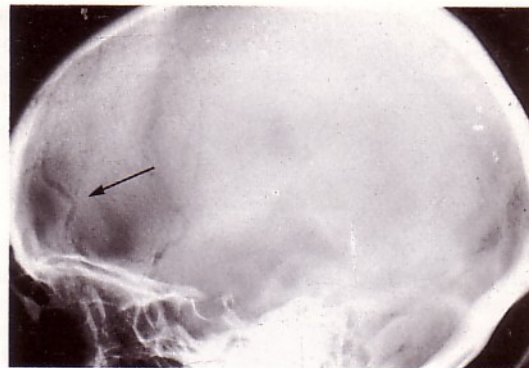
■ Skull film



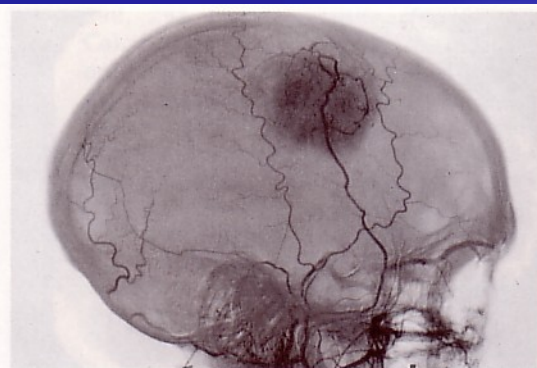
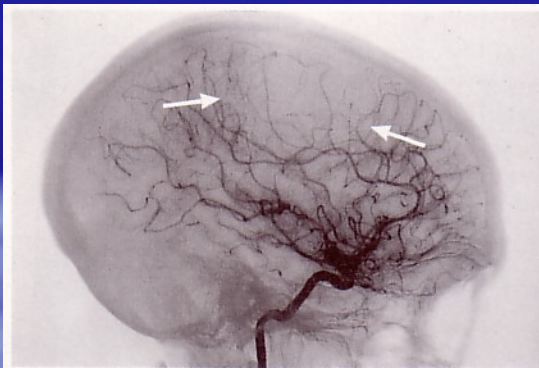
290, 291 Calcification in meningiomas. Calcification, which is common, may be visible on the plain radiograph. This convexity meningioma was found by accident when the (asymptomatic) patient was X-rayed after a minor road traffic accident (290, arrowed). A similar appearance produced by an Asian patient's thick plaits (291, arrowed) shows the importance of establishing that the 'lesion' is intracranial.



292 Hyperostosis due to meningioma. Seven years before admission this 81-year-old man had an attack of shingles in the first division of the right trigeminus. Subsequently he became forgetful, disorientated and occasionally incontinent of urine. Nine months before admission he began to have fits. Examination confirmed that he was mildly demented but there were no other relevant findings. The skull radiograph showed hyperostosis of the sphenoid bone (292, arrowed), which was much more evident on the scan, on which a large enhancing lesion surrounded by oedema was visible. The patient was unfit for surgery and a post-mortem 6 months later confirmed that the lesion was a meningioma (288, see also 77)



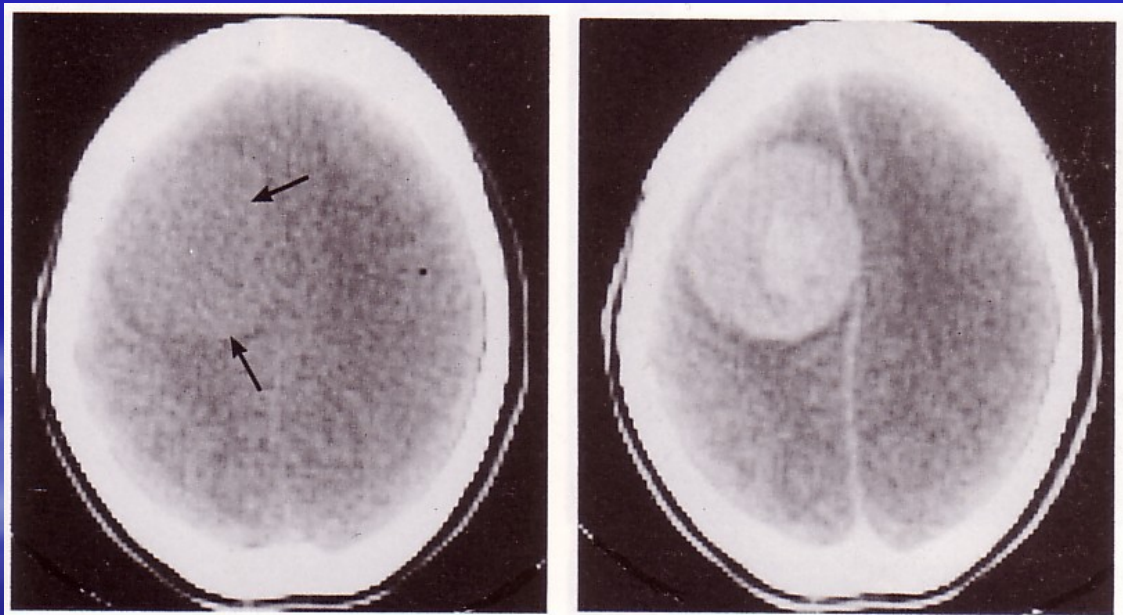
293, 294 Parasagittal meningioma with vascular markings. Prominent vascular markings on the left side of the skull of a 38-year-old woman with chronic papilloedema (293, arrowed). She gave a 6-month history of transient blurring of vision on exertion, mild headaches and occasional dizziness on bending. The isotope scan, a very sensitive technique for demonstrating meningiomas, showed a parasagittal lesion in the left frontal region (294, arrowed).



297, 298 Angiogram of meningioma. Angiography characteristically shows displacement of vessels and a blush. On this internal carotid arteriogram the position of a large tumour is marked by stretched vessels surrounding an avascular area (297, arrowed). The blood supply, however, came from the external carotid and the tumour blush only appears when that vessel is injected (298).

Imaging of meningioma

- CT scan has homogenous enhancement and dural tail sign



295, 296 CT scan of meningioma. On a CT scan, a meningioma usually appears as a well defined mass adjacent to bone or to the falx (295, arrowed) which enhances vigorously with contrast (296).

- MRI show well circumscribe mass, homogenous enhancement, hypointensity in T1 and T2

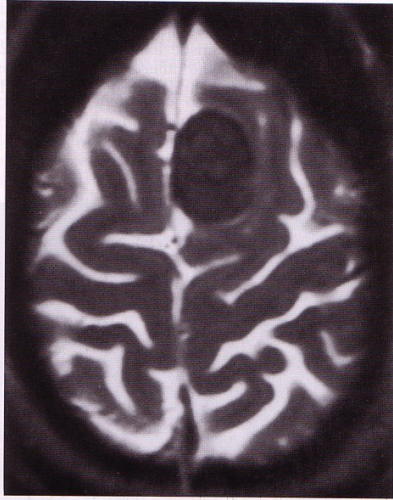
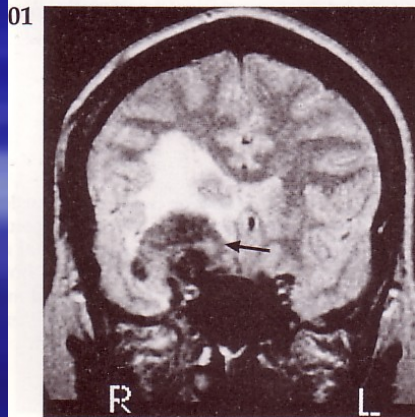
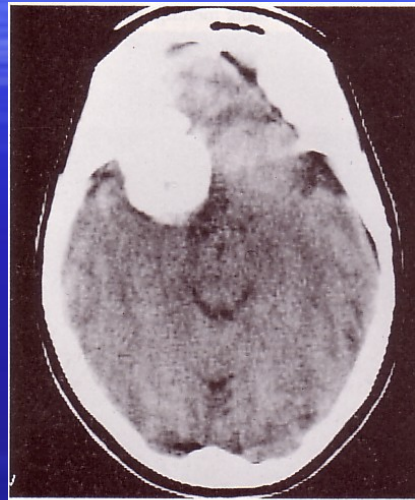


FIGURE 2-2 MENINGIOMA

Unlike most intracranial tumors, which are hyperintense, or white, on T2-weighted MR images, meningiomas are isointense, if not hypointense, when compared with cortical gray matter. This fibrous meningioma was exceptionally dark because of its content of collagen and calcium.



299-301 MR scan of meningioma. A sphenoid wing meningioma seen for comparison on a CT scan (299) and on axial (300) and frontal (301) MR scans. The white area surrounding the tumour on the MR scans is oedema. Meningiomas may be overlooked on an MR scan if contrast is not given.

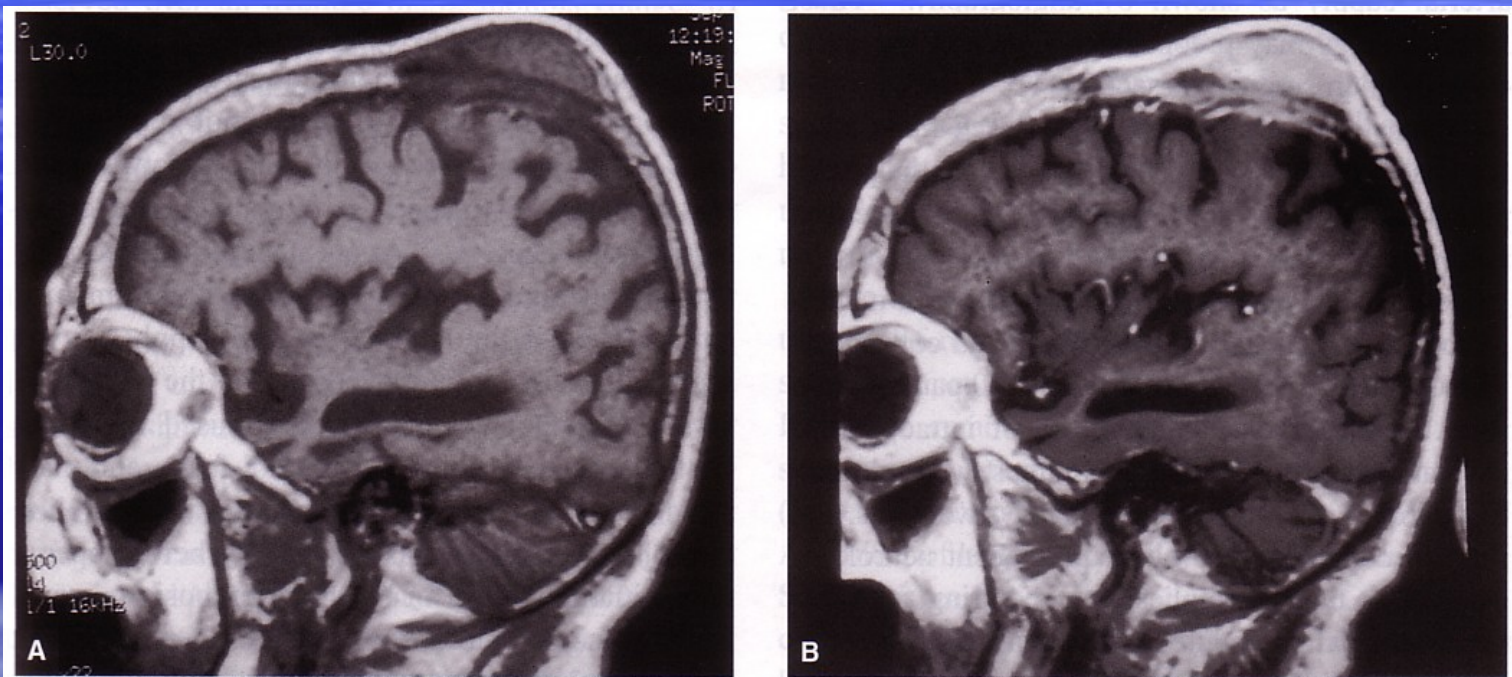


Figure 18-16. A, Sagittal T1-weighted magnetic resonance imaging without contrast shows a loss of diploic-space signal hyperintensity due to invasion by meningioma. B, Postcontrast image confirms the enhancement of tumor in bone producing a "tower skull." Surgical excision specimen is the same as that seen in Figure 18-2.

Treatment of meningioma

- Surgery is main therapy

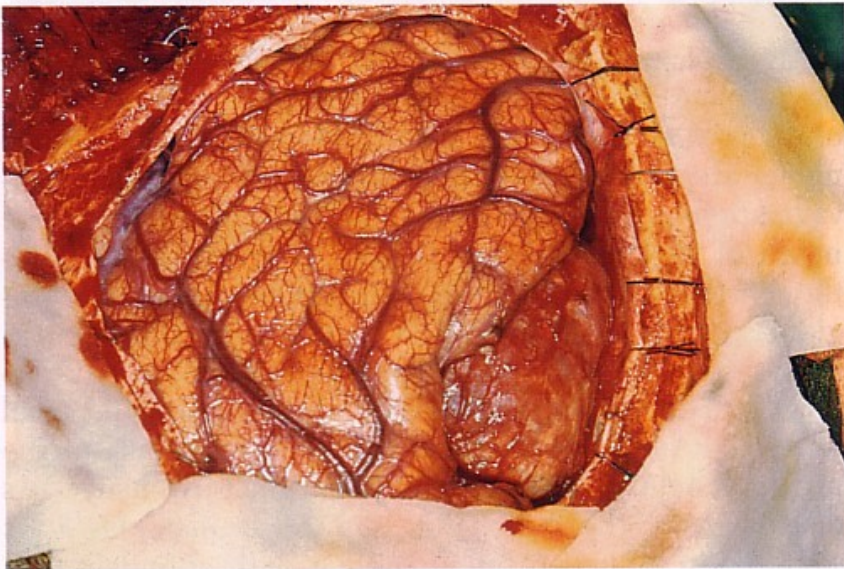


FIGURE 2-10 MENINGIOMA

The typical extra-axial position of most meningiomas is readily evident at surgery.

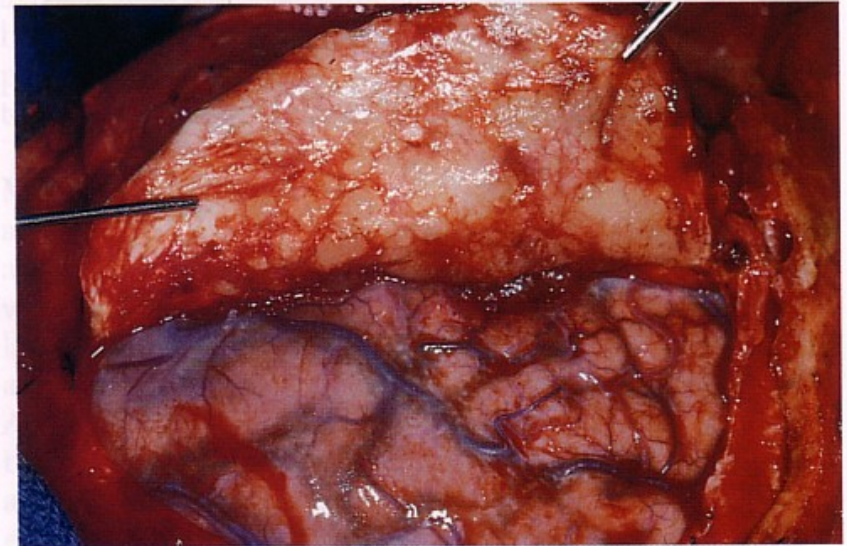


FIGURE 2-11 MENINGIOMA EN PLAQUE

Occurring most often in the temporal region and covering the sphenoid wing, *en plaque* meningiomas spread horizontally without the formation of a globular mass. (Courtesy of Dr. John C. D. Atkinson, Rochester, MN.)

Intraoperative

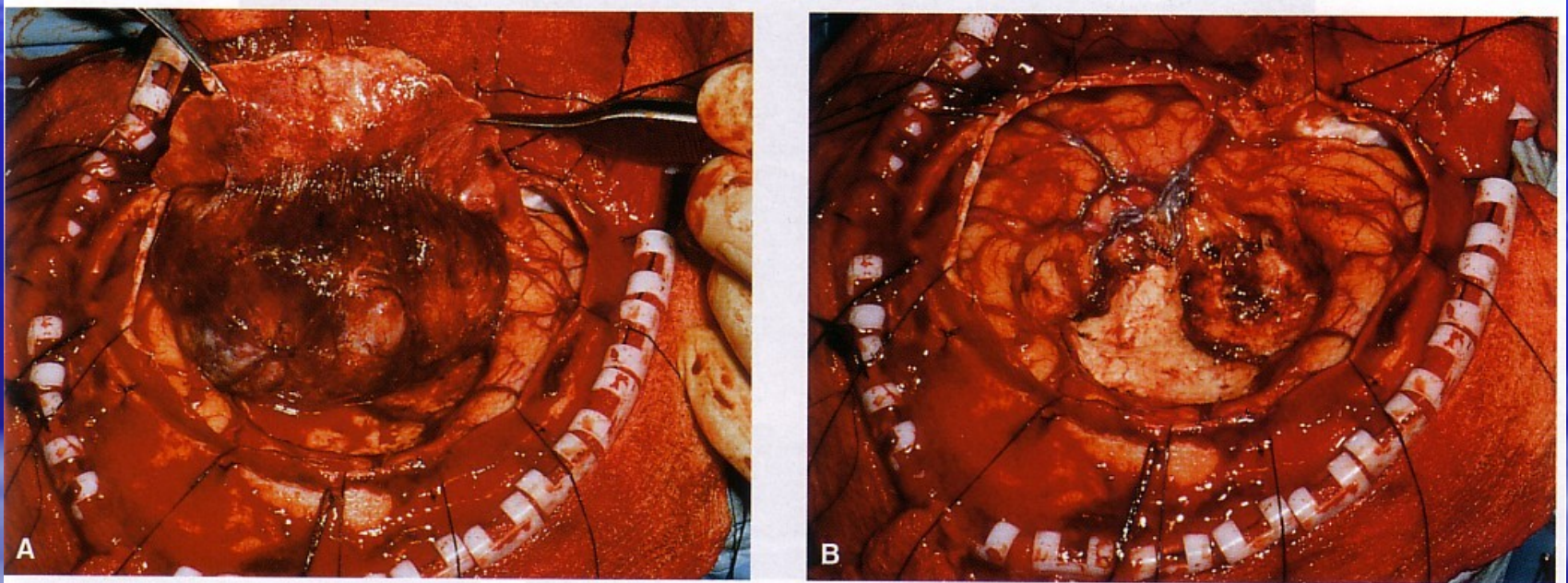


Figure 18–11. A, Right frontal temporal, parietal convexity meningioma being removed with a cuff of hypervascular dura that was consistent on MRI, with a “dural tail.” B, Tumor has been removed (gross specimen is seen in Figure 18–1), with preservation of all but the most medial arachnoid plane. The white tissue in the depths of the cavity is subcortical white matter.



Figure 18-10. A, Axial magnetic resonance imaging (MRI) scan of a large left-sided convexity meningioma. B, Coronal scan of same tumor. The patient presented with complex language disturbance and with blurred vision from hemorrhagic papilledema. On the axial scan, note the edema at the deepest tumor-brain interface, where at surgery, the arachnoid plane was absent. C, Postoperative axial MRI with contrast. The patient was discharged on the third postoperative day.

Gross pathology



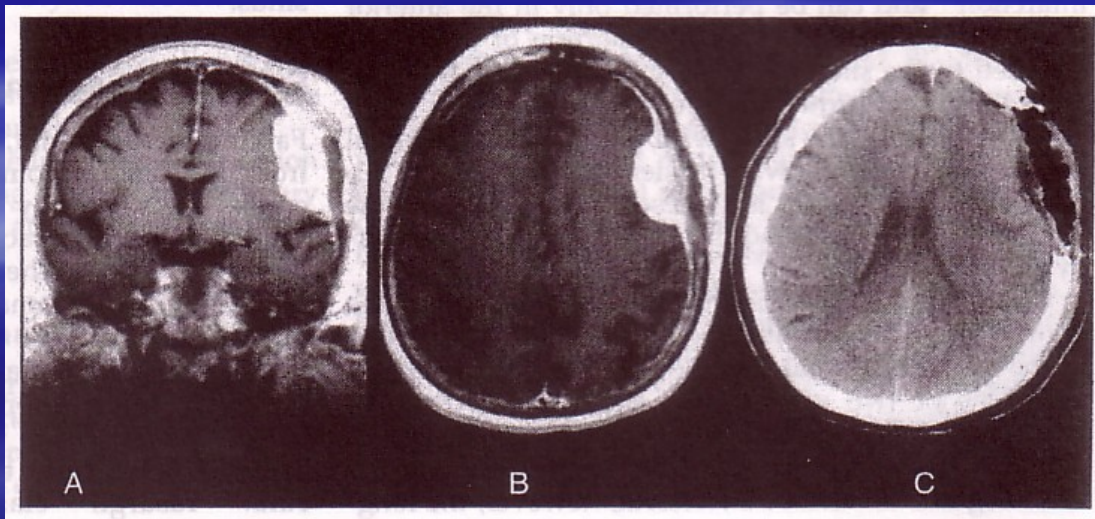
FIGURE 2-12 MENINGIOMA

Meningiomas often appear lobular on cut section.



FIGURE 2-13 MENINGIOMA

Fibrous meningiomas are often remarkably smooth surfaced.



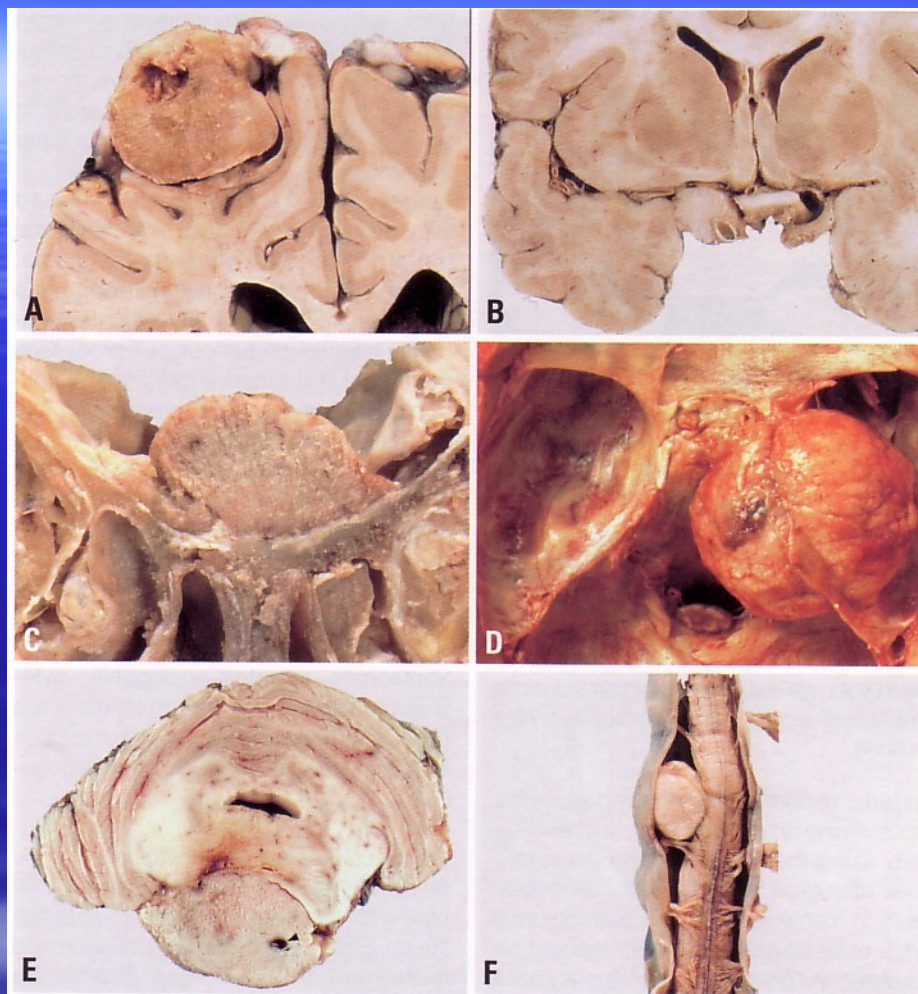
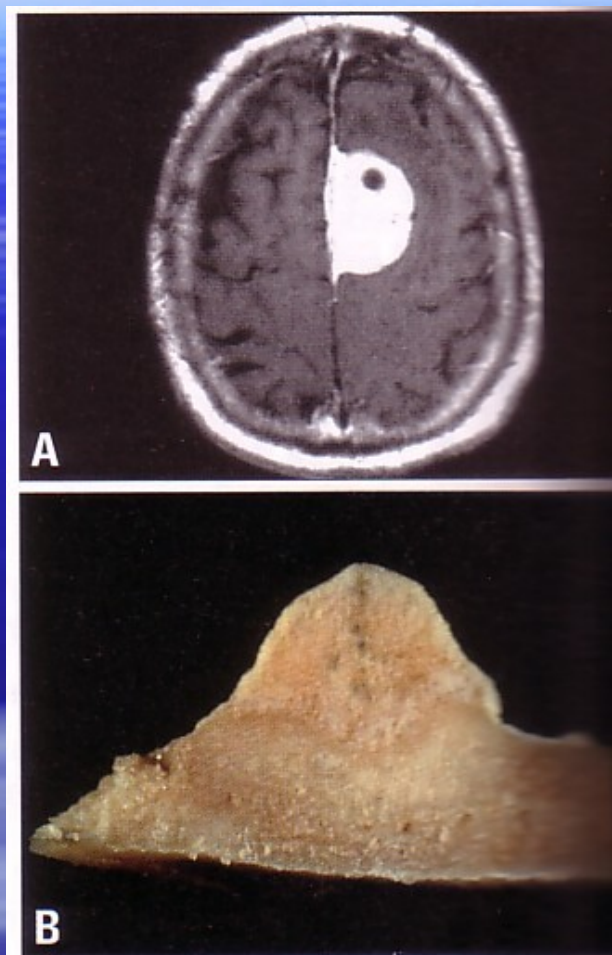


Fig. 11.3 Macroscopic features of meningioma. **A** Large parasagittal meningioma compressing the adjacent parietal lobe. **B** Meningioma of the medial sphenoid wing encasing the carotid artery. **C** Coronal section through the anterior cranial fossa showing a meningioma of the olfactory groove. **D** Meningioma of the sphenoid wing. **E** Large meningioma of the clivus compressing the brain stem. **F** Spinal meningioma compressing the spinal cord.

Extent of surgery

TABLE 62-5 ■ Simpson Grade

Grade I	Macroscopically complete tumor removal with excision of the tumor's dural attachment and any abnormal bone
Grade II	Macroscopically complete tumor removal with coagulation of its dural attachment
Grade III	Macroscopically complete removal of the intradural tumor without resection or coagulation of its dural attachment or extradural extensions
Grade IV	Subtotal removal of the tumor
Grade V	Simple decompression of the tumor

Adapted from Simpson D: The recurrence of intracranial meningiomas after surgical treatment. *J Neurol Neurosurg Psychiatry* 20:22, 1957.

Table 18-4. SIMPSON CLASSIFICATION OF TUMOR GRADE BY EXTENT OF EXCISION OF TUMOR, DURA, AND BONE OR VENOUS SINUS

Grade	Tumor Removal			Dural Attachment			Bone / Sinus Excised	
	Complete	Partial	Biopsy	Excised	Coagulated	Neither	Yes	No
I	X	—	—	X	—	—	X	—
II	X	—	—	—	X	—	X	—
III	X	—	—	—	—	X	—	X
IV	—	X	—	—	—	X	—	X
V	—	—	X	—	—	X	—	X

Treatment of meningioma

- **Radiosurgery** can do in case of residual of tumour after operation, size less than 3 cm in diameter and limited at optic apparatus

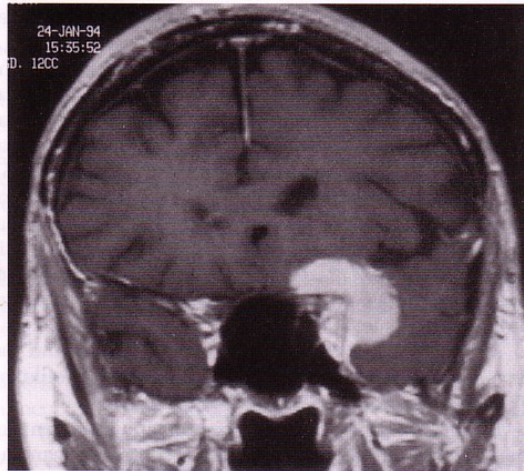


FIGURE 2-3 MENINGIOMA

Meningiomas are contrast enhancing, with a broad dural base and short extensions of enhancement along the dura ("dural tail" sign). A "tail" extends inferiorly from this sphenoid ridge tumor.



FIGURE 2-5 MENINGIOMA OF THE OPTIC NERVE SHEATH

Exophthalmos and visual loss were clinical expressions of this tumor situated immediately behind the globe of a 46-year-old woman. The discrete neoplasm expands within the optic sheath at the expense of the inferiorly displaced nerve.

- Optic nerve sheath meningioma

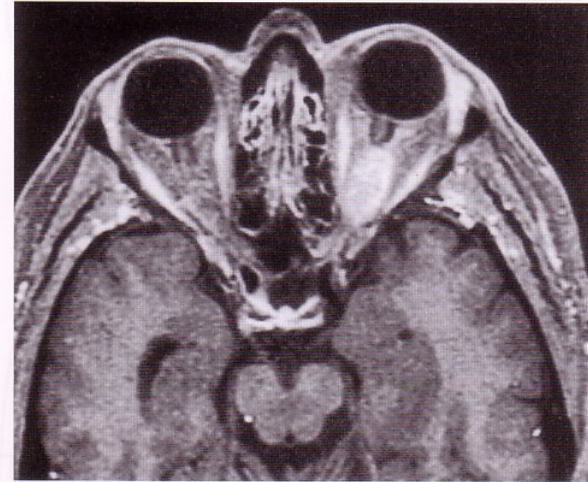
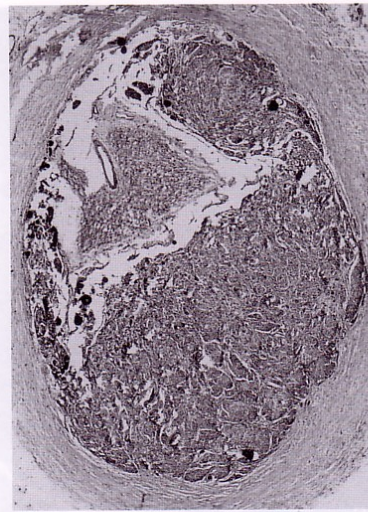


FIGURE 2-4 MENINGIOMA OF THE OPTIC NERVE SHEATH

Exophthalmos and visual loss drew attention to this contrast-enhancing mass that encased the optic nerve and displaced the globe anteriorly.



2-6



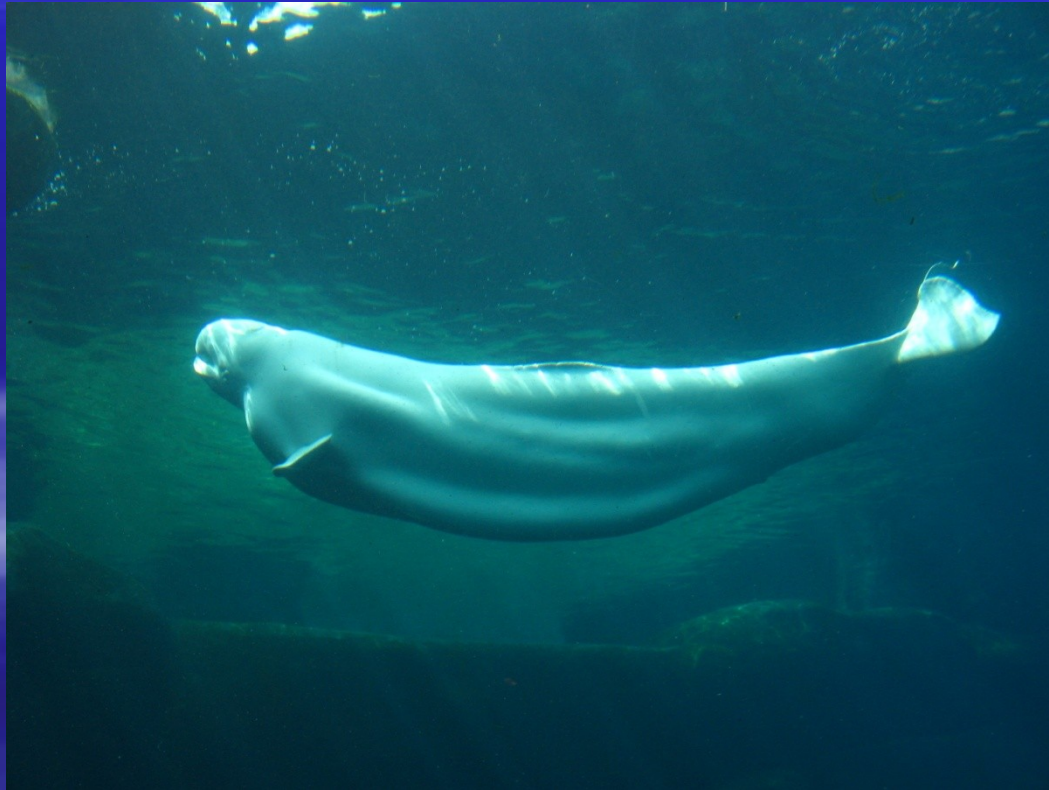
2-7

FIGURES 2-6 AND 2-7 MENINGIOMA OF THE OPTIC NERVE SHEATH

Confined by the dura, the neoplasm in Figure 2-6 compressed the optic nerve into the triangular structure that is visible at the top of the illustration. Meningiomas in this location may also freely infiltrate the dura and extend outward into orbital soft tissues (Fig. 2-7).

prognosis

- Depend on extent of surgery



Acoustic neuroma



FIGURE 12-24 SCHWANNOMA

Externally and on cut section, the gross appearance of schwannomas varies, with some tumors being multinodular (A). On cut section, they may be solid and yellow-tan in color (B), necrotic and hemorrhagic with cystic change (C), or thin walled and almost entirely cystic (D).

TABLE 12-2**Clinicopathologic Comparison of Schwannoma and Neurofibroma**

Schwannoma	Neurofibroma*
Non-NF1 associated/occasional NF2 or schwannomatosis associated	NF1 associated/non-NF2 associated
Frequently affects extremities	Frequently involves the trunk
Usually solitary	Frequently solitary
Nerve often identified	Nerve infrequently identified
Eccentric to nerve	Incorporates nerve
Globular or plexiform	Fusiform, globular, plexiform, or diffuse
Encapsulated	Delicately surrounded by perineurium and epineurium (solitary and plexiform neurofibroma); no capsule (diffuse neurofibroma)
Nonmucoid, soft to firm	Mucoid and firm
Tan to yellow, opaque	Gray-tan, opalescent
High cellularity	Low to moderate cellularity
Biphasic Antoni A and B patterns	Uniphasic pattern with gradual changes in cellularity
Scant or no stromal mucin	Mucin-rich matrix
Axons often absent	Axons often present
Palisading and Verocay bodies	Wagner-Meissner or rarely pacinian-like corpuscles; no palisading
Composed of Schwann cells	Composed of Schwann cells, perineurial-like cells, fibroblasts, and transitional cells
Mast cells infrequent	Mast cells frequent
Malignant transformation extremely rare	Malignant transformation rare (2% of patients with NF1)

*Exclusive of plexiform neurofibroma.

NF, neurofibromatosis.

Clinical manefestation

- Tinnitus
- Hearing loss on side of tumour
- Ataxia
- Facial numbness
- Diplopia
- Hemiplegia
- Facial palsy
- Slow progressive

Imaging

- CT scan can show widening of IAC and heterogenous mass at CP angle

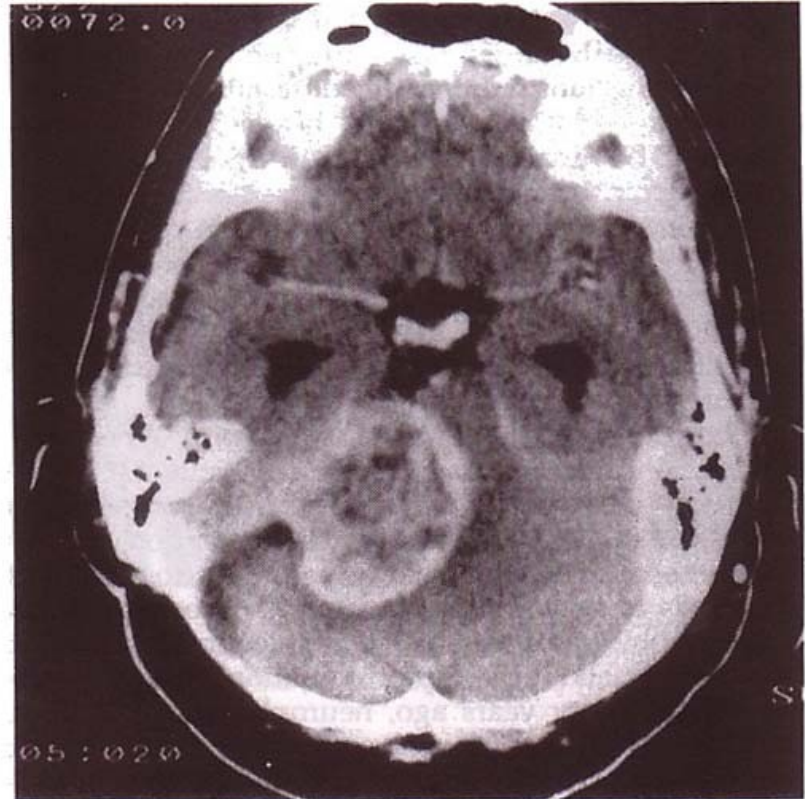
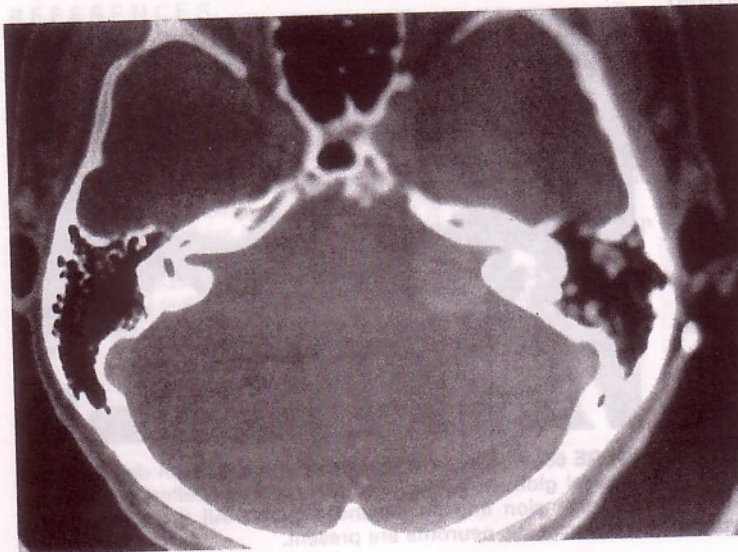
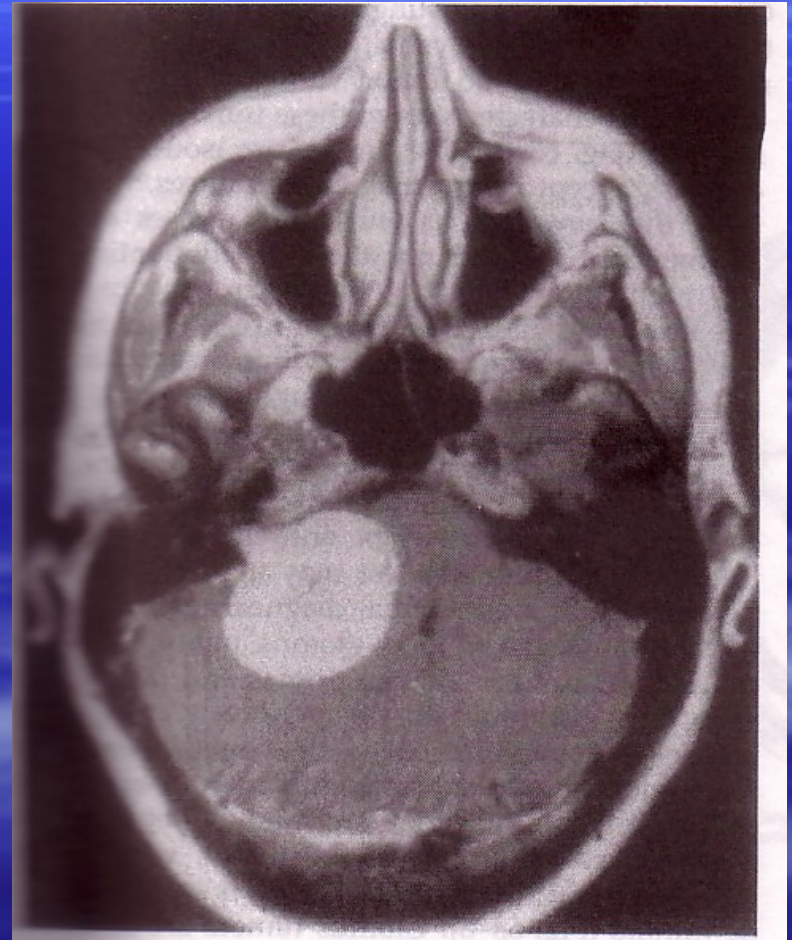


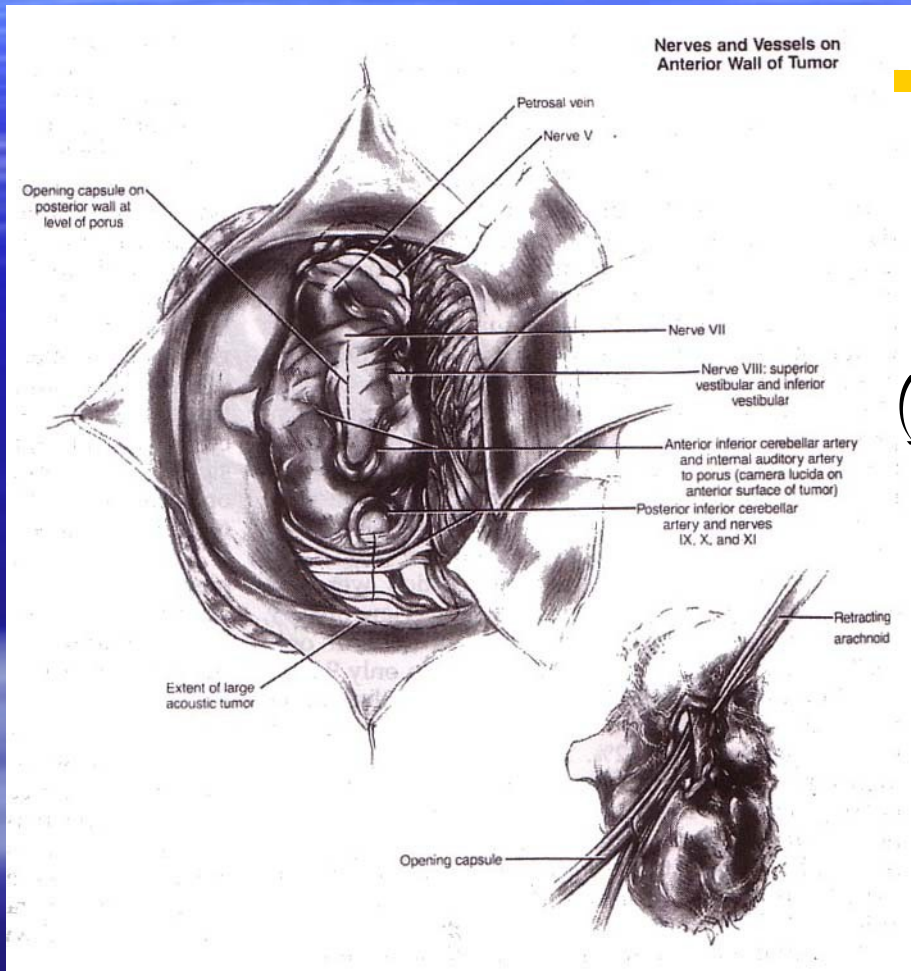
FIGURE 65-6. This roughly spherical tumor demonstrates a broad base, with diffuse involvement of the petrous bone. The broad base differentiates this meningioma from an acoustic neuroma.

Imaging of acoustic tumour

- MRI show typical spherical tumour, erosion and extension of tumour into the porus acusticus



Surgical treatment



■ Surgical approach
most case use
suboccipital craniectomy
(retrosigmoid approach)

Treatment of acoustic tumour

- Radiosurgery

limit tumour size less than 2.5 cm.

after radiation ,tumour still persist in scan about 50%

Technique of neurosurgery

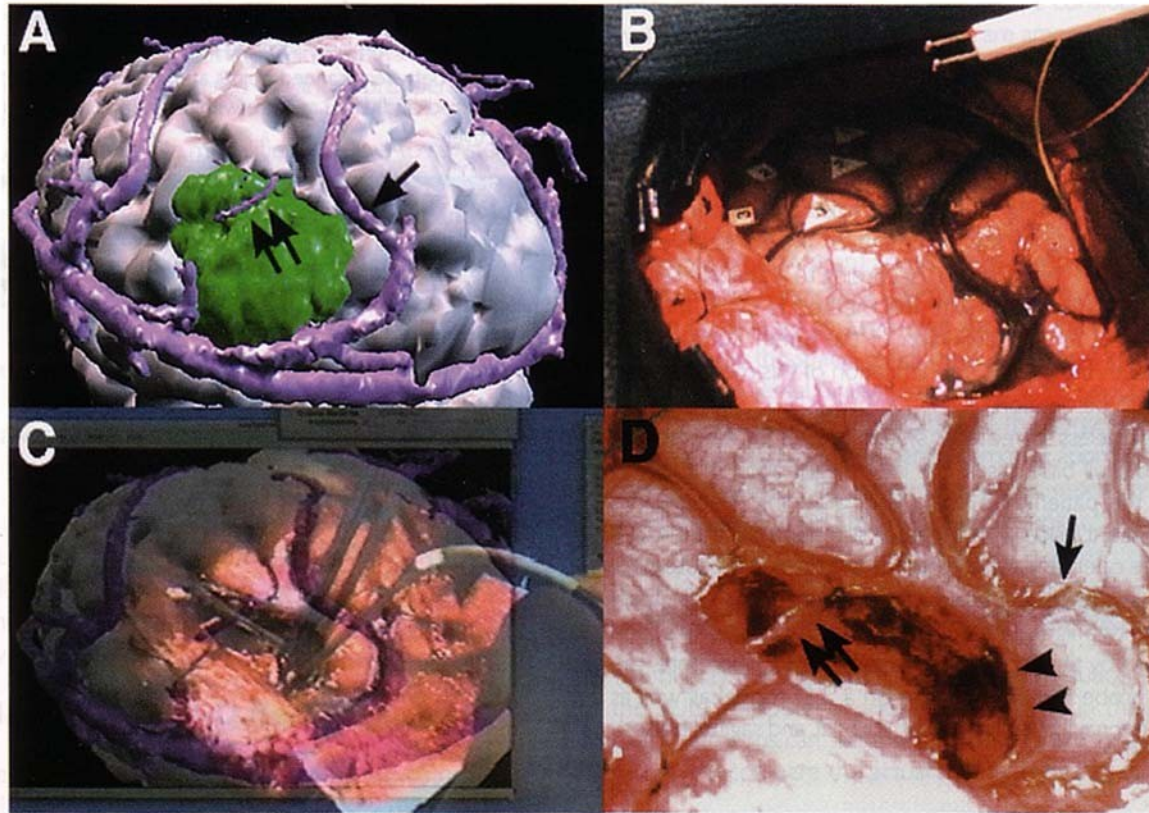
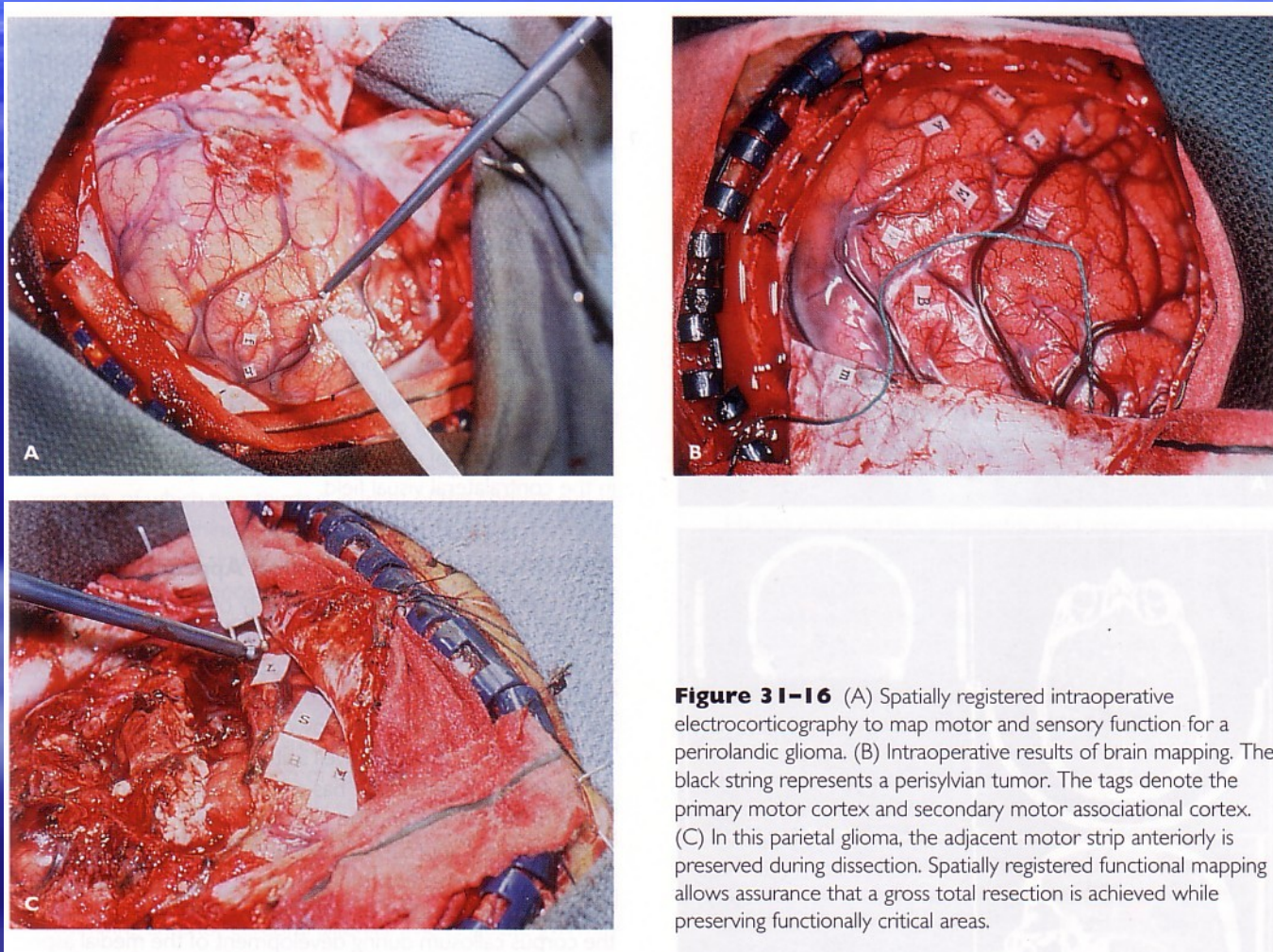


Figure 11-3 An illustrative case of vessel-to-vessel registration. (A) The 3-D model display consisting of the brain, cortical vessels, and tumor. The tumor is colored in green and the vessels are colored in purple. The upper side of the image is the patient's right side and the left side is the patient's anterior side. There are two large cortical vessels (single arrow and double arrows). (B) The surgical field in the same alignment as the 3-D model. The two large cortical vessels were easily identified on the surgical field. The primary motor cortex (squares) and the primary sensory cortex (triangles) were confirmed by cortical mapping. (C) The 3-D model display was superimposed onto the video image of the actual surgical field. (D) Gross total resection was achieved with the help of video registration. One of the large cortical vessels (double arrows) was preserved on the resected cavity (arrow heads). The other large cortical vessel is consistent with that of the 3-D model display (single arrow). Note that the sulci are difficult to identify because they are hidden under the vascular network.

Cortical stimulation

- Help to locate functional area language, motor area etc.

Cortical and subcortical stimulation



Navigation can reduce unnecessary surgical incision

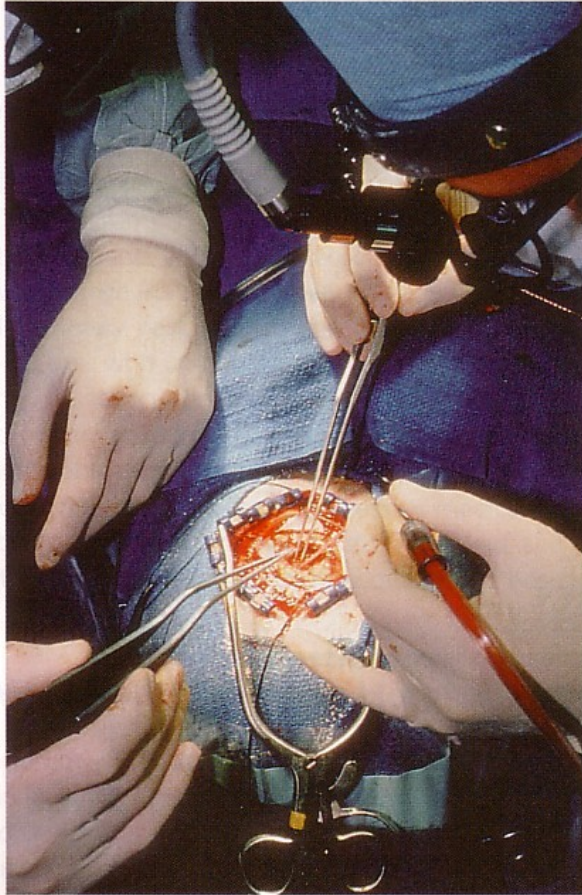
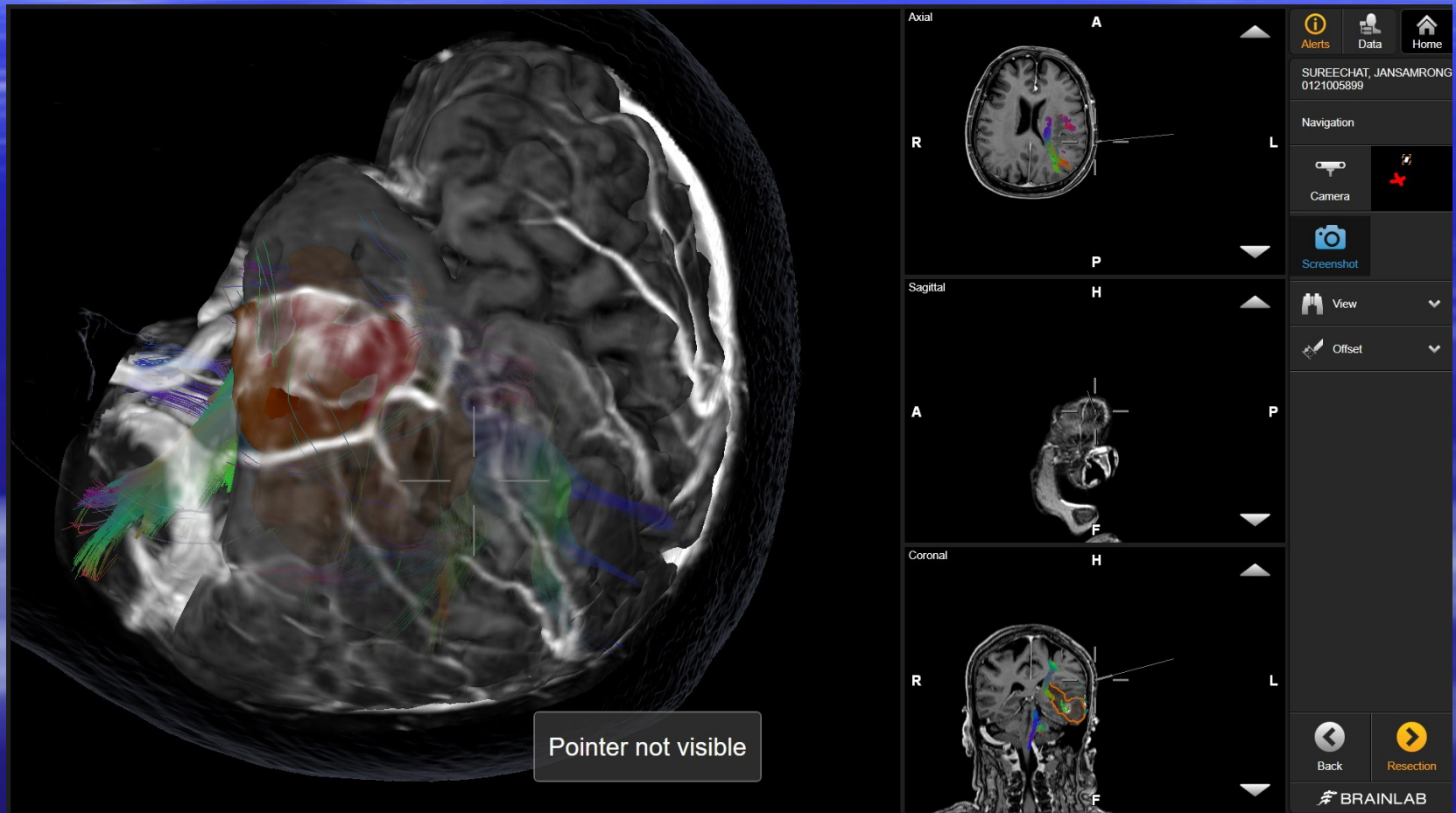


Figure 22-9 Resection of an intracerebral metastasis via a small linear scalp incision and 1-in trephine craniotomy allows minimization of the invasiveness of the procedure and a more rapid

Image fusion with MRI and functional area plus fiber tracking

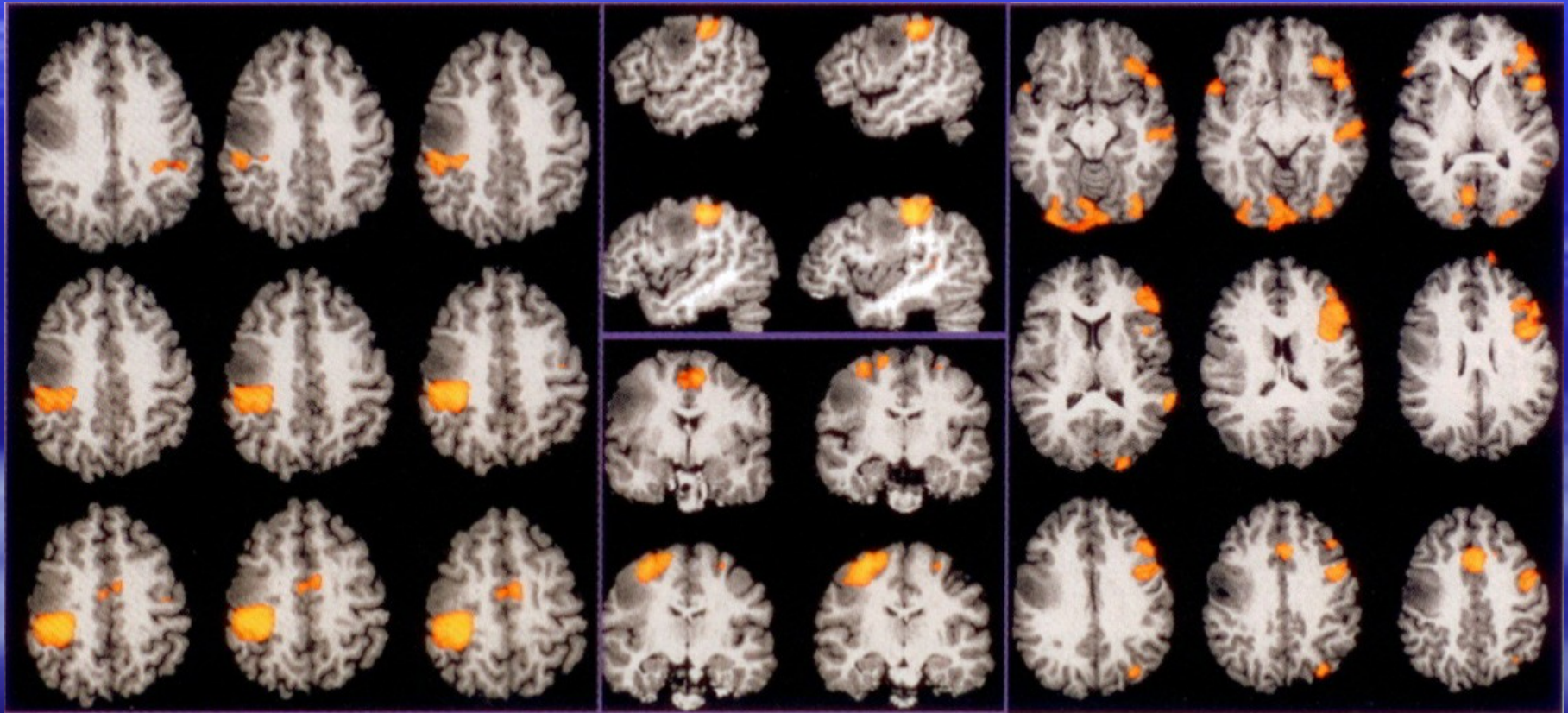




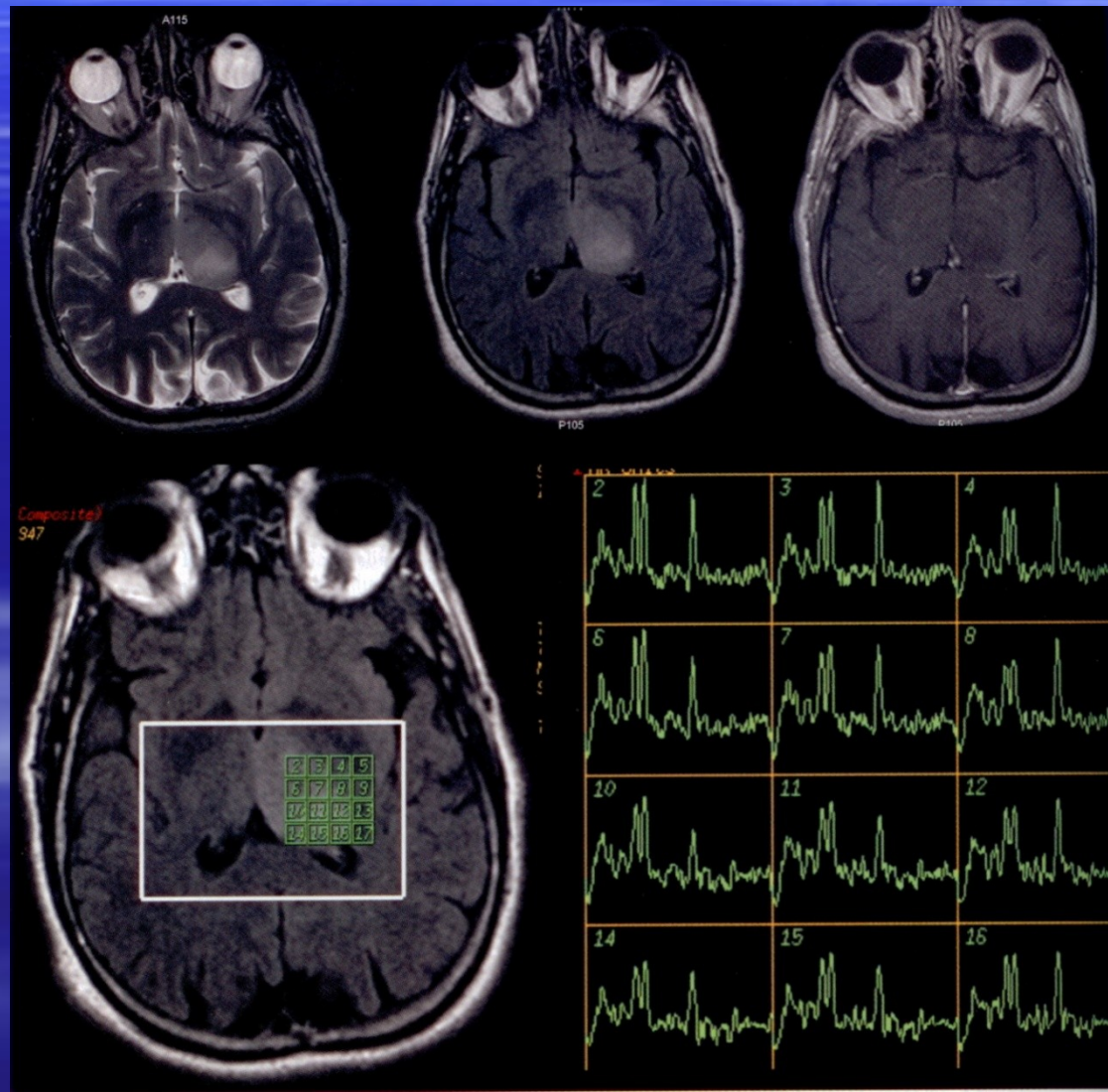
Modern Neuromicroscope



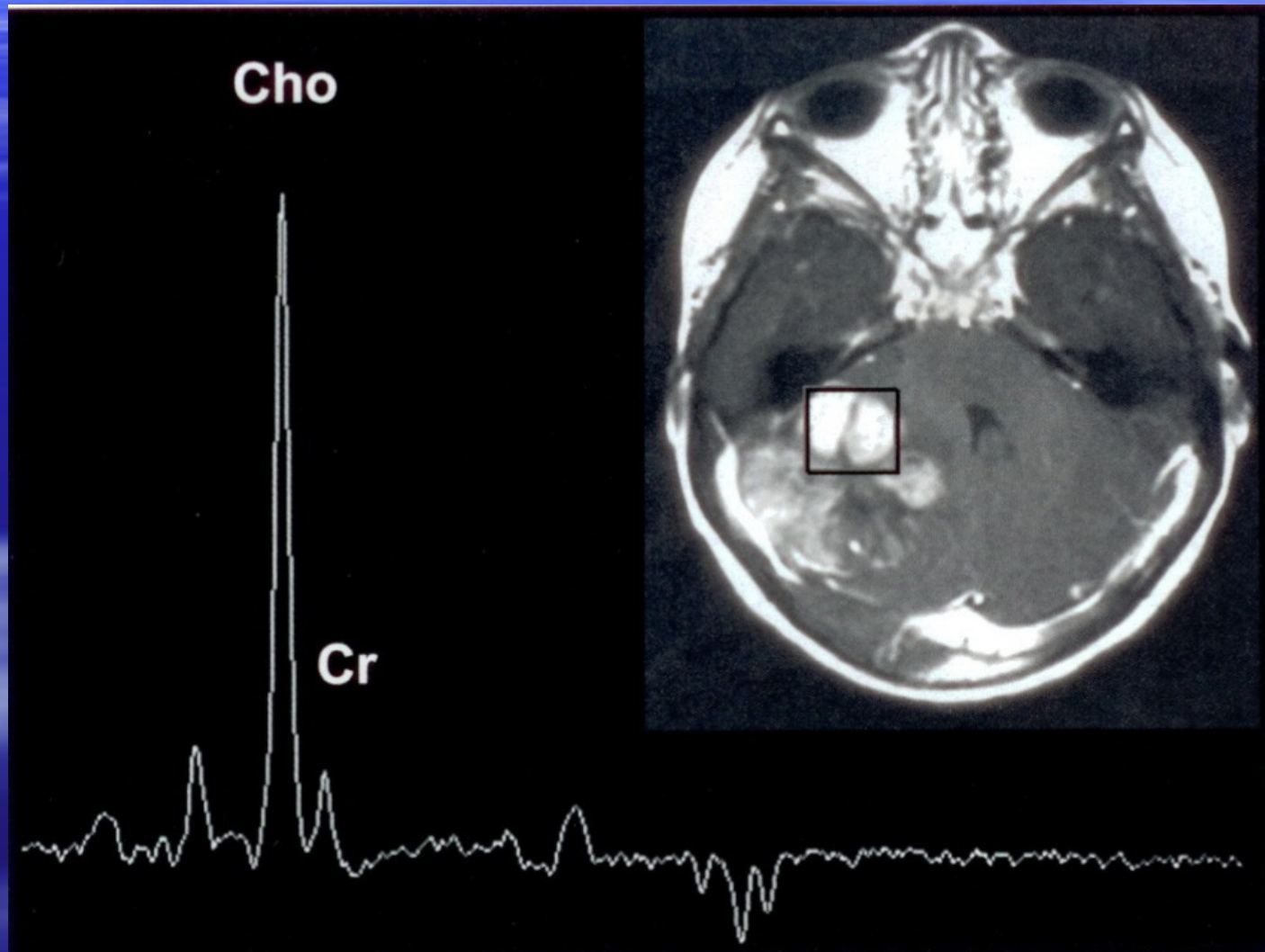
Functional MRI



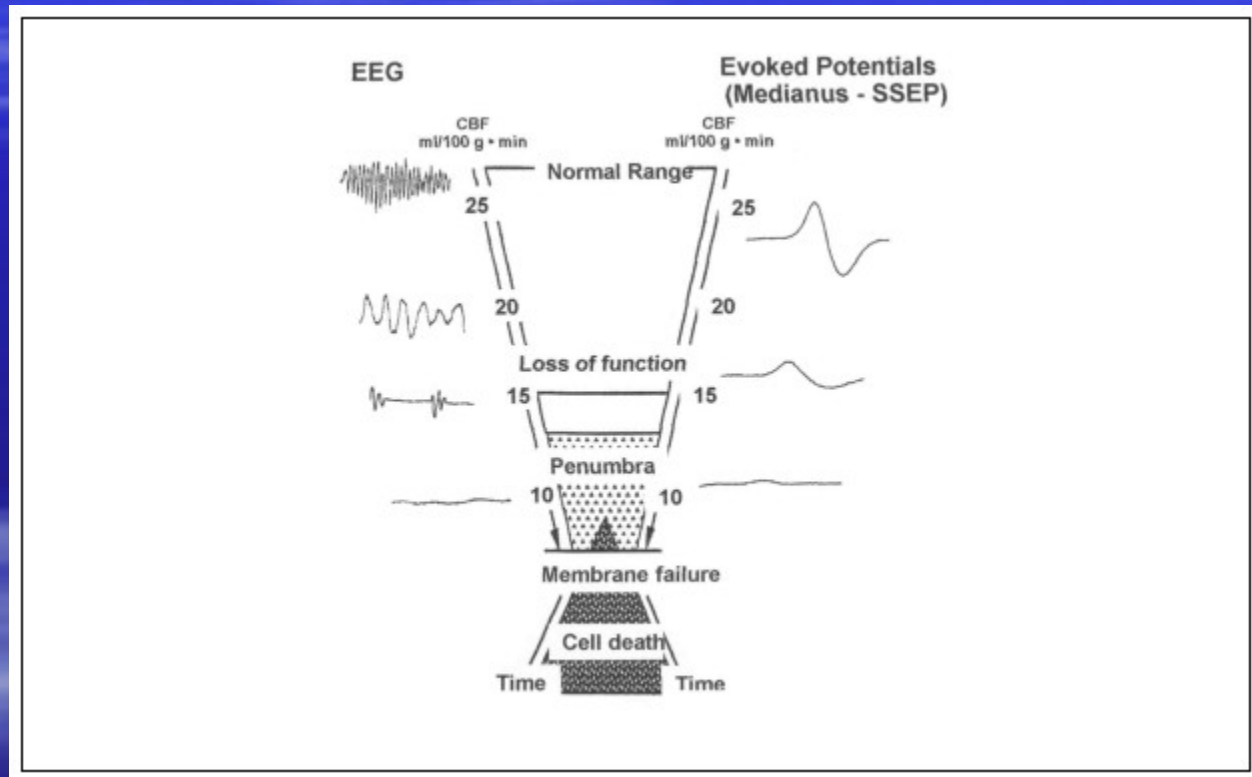
Magnetic Resonance spectroscopy(MRS)



MRS



Neurophysiologic monitoring



Portable CT Scan



Excellent result

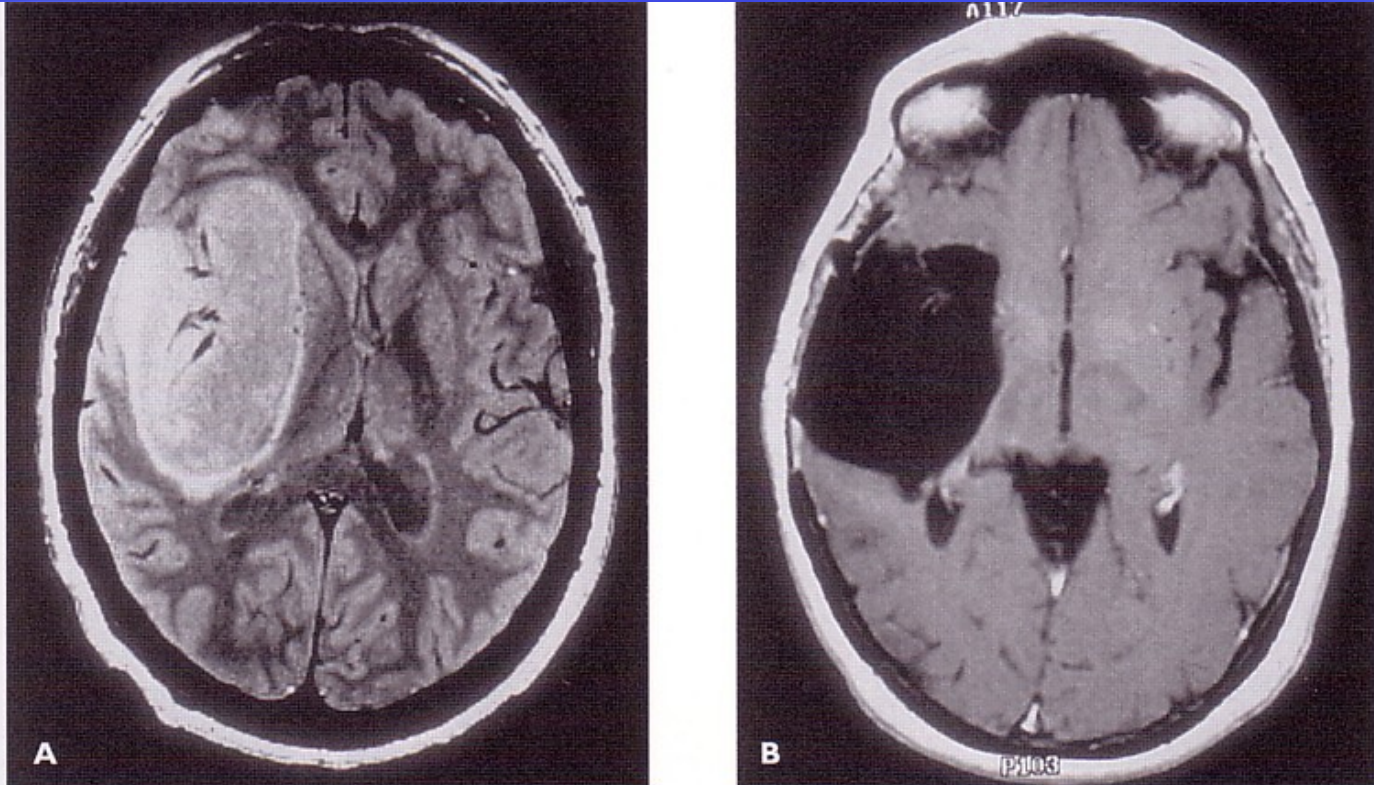


Figure 31-22 (A) Preoperative proton density image demonstrating a temporal/subinsular oligodendroglioma. (B) Postoperative image demonstrating a gross total resection of the tumor as deemed by postoperative imaging criteria. The patient was neurologically intact.

Stereotactic biopsy

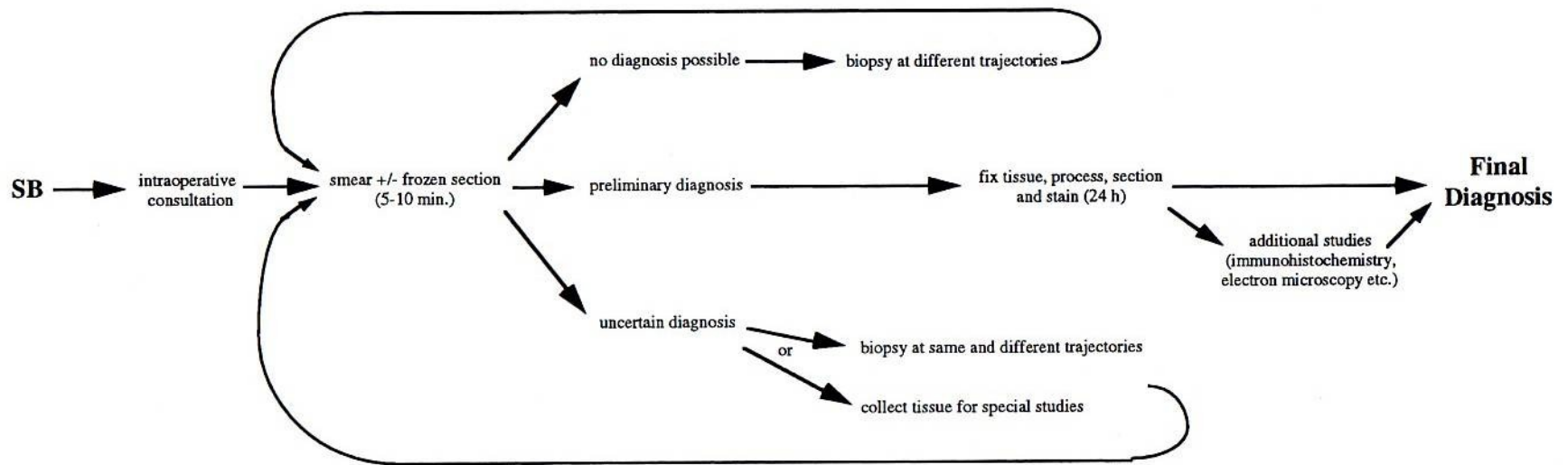


Diagram 16-1 Approach to analysis of stereotactic brain biopsy. This flow chart outlines a typical approach to the analysis of tissue from a brain biopsy. The times and methods will vary depending on the institution and nature of the patient's disease.

Stereotactic frame

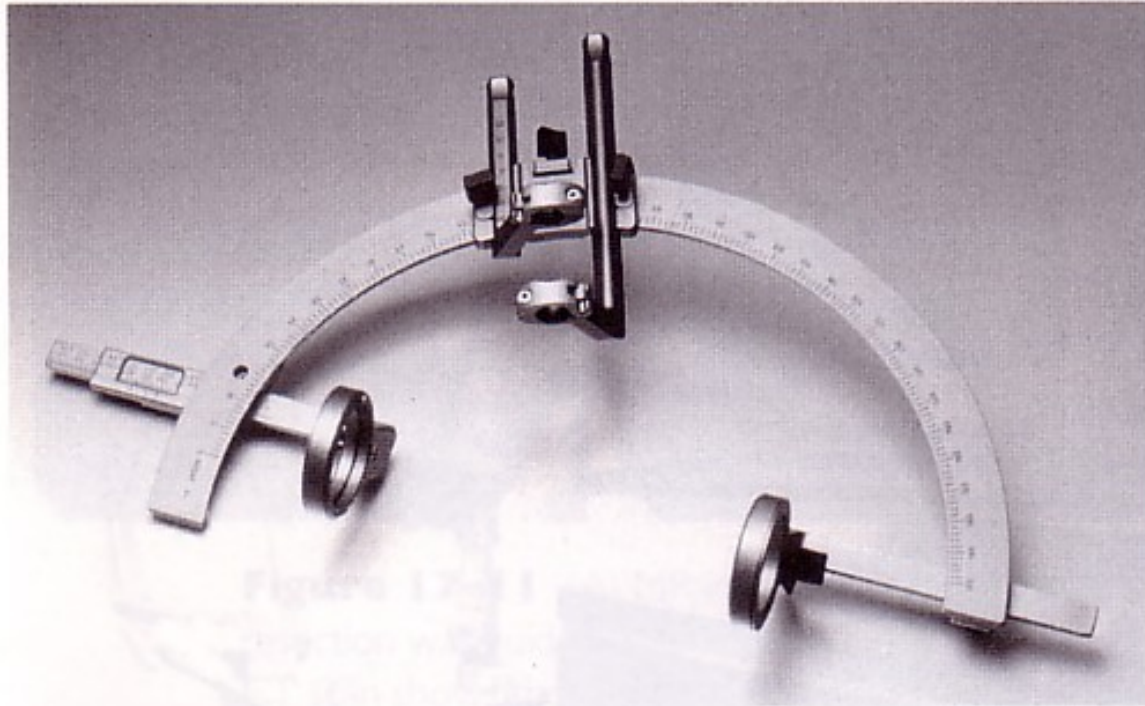


Figure 17-7 The Leksell Model G expanded stereotactic arc can be attached on the y or x frame component.

Stereotactic instrument

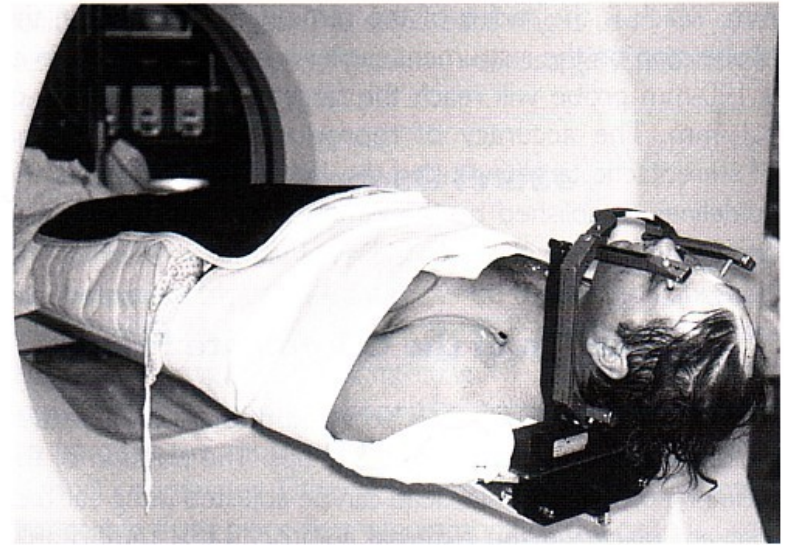
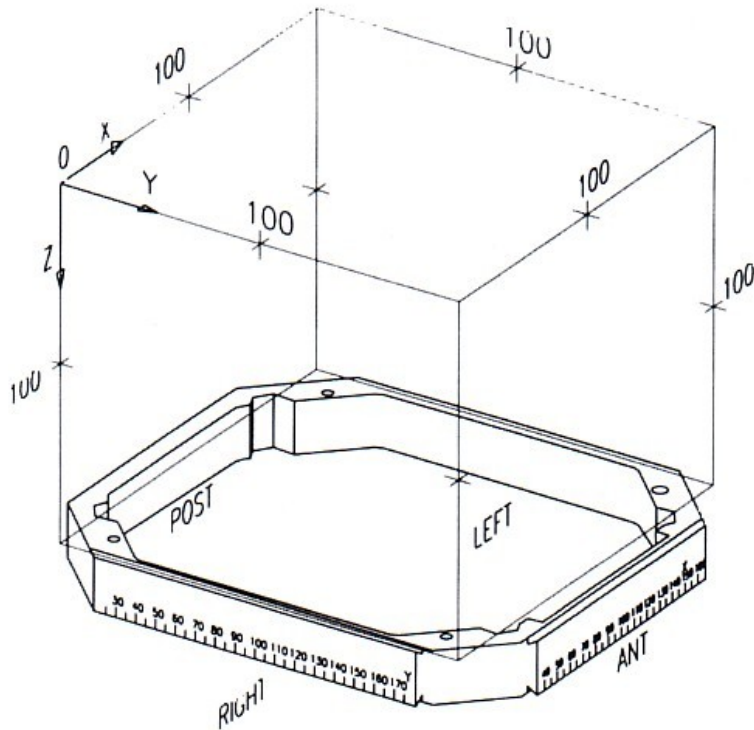


Figure 17-6 The patient in position for CT localization. The fiducial system can be seen attached to the base ring.

Biopsy set

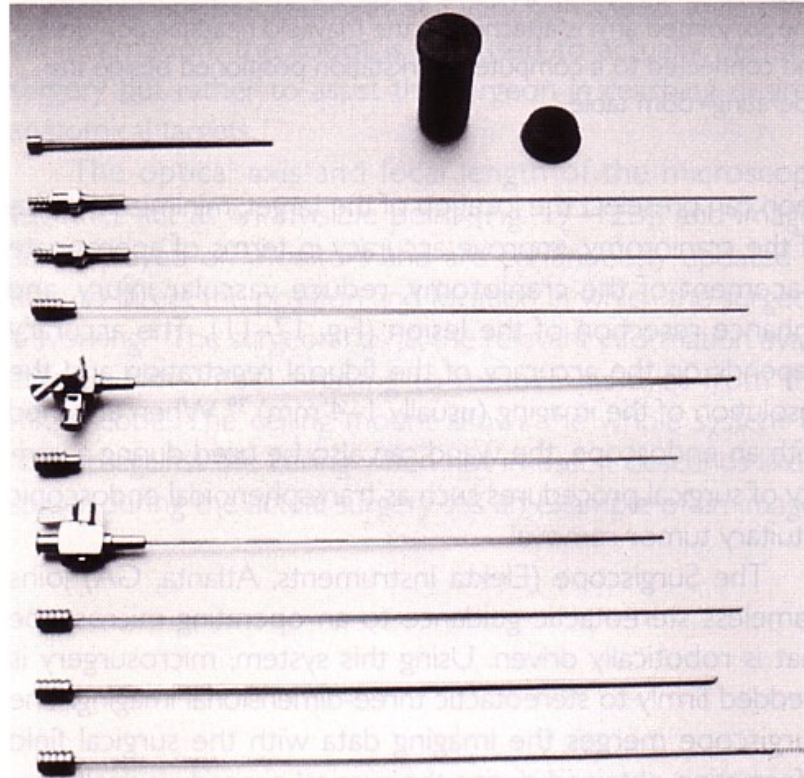


Figure 17-8 The Backlund biopsy set includes both cyst puncture needles and the "spiral" biopsy system. Most biopsies are performed with a side-coring instrument.

Computer work station

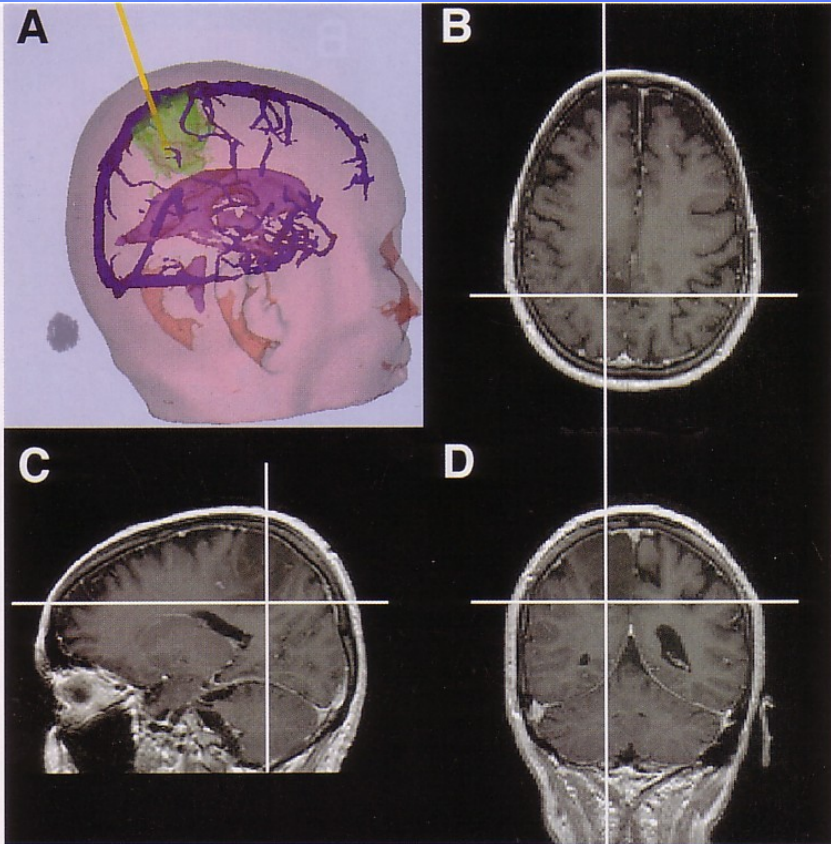


Figure 11-5 The interface of the surgical navigator. The surgeon confirmed the position of the bottom of the tumor. (A) The tip of the probe (yellow line) is displayed on the 3-D model which shows the tumor colored in green. (B–D) The white cross shows the tip of the probe on the three orthogonal MR images simultaneously.

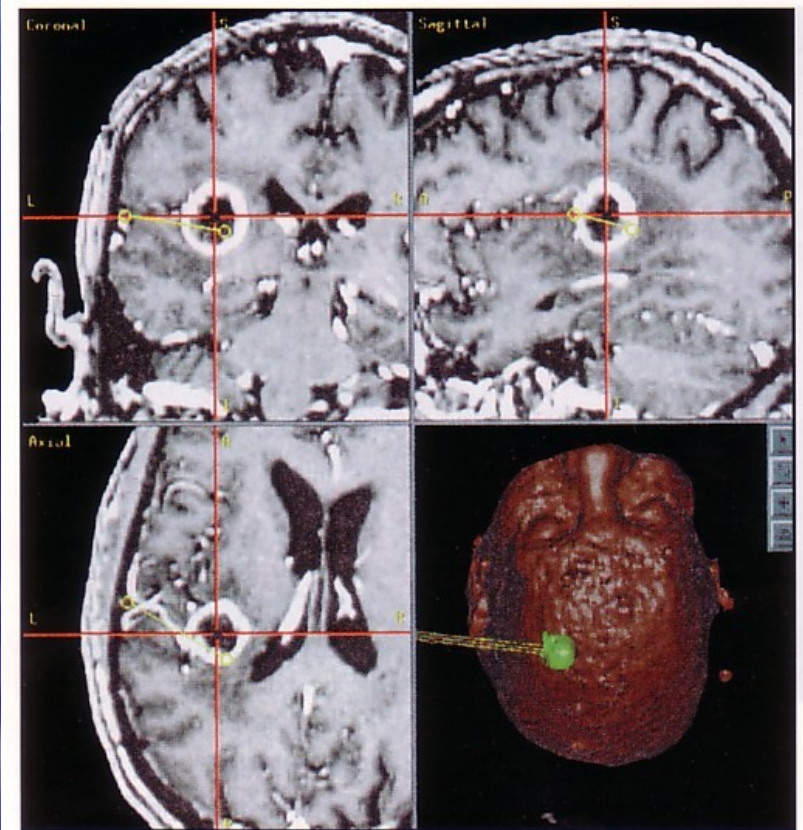


Figure 28-9 Imbedded tumor viewed by surface rendering.

Neuronavigation in Siriraj hospital



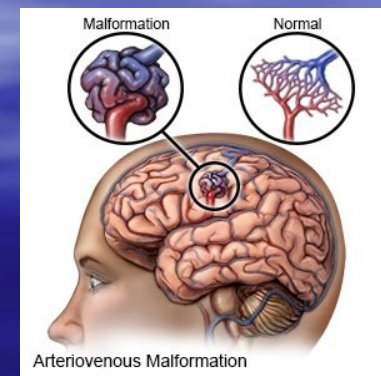
Postop care in brain tumor surgery

- Control blood pressure
- Avoid acidosis
- Hypoxia
- Prevention aspiration
- Rebleeding after surgery recognition
- Avoid seizure



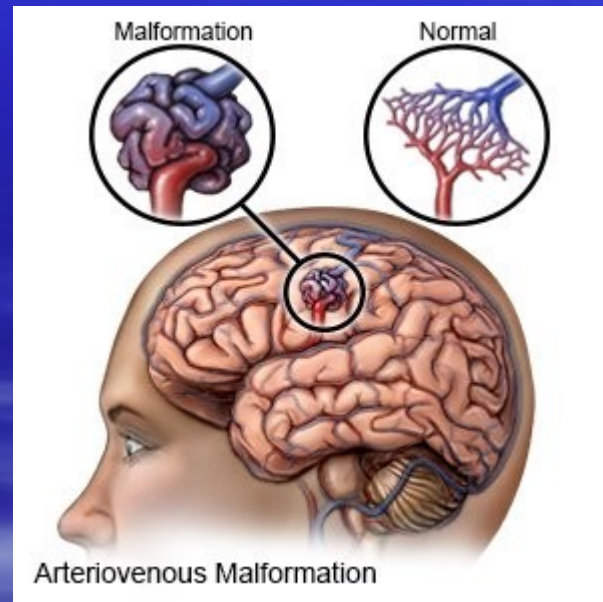
Cerebral arteriovenous malformation(AVM)

- Subtype
 - arteriovenous malformation
 - cavernous malformation
 - capillary telangiectasia
 - developmental venous anomaly(DVA)



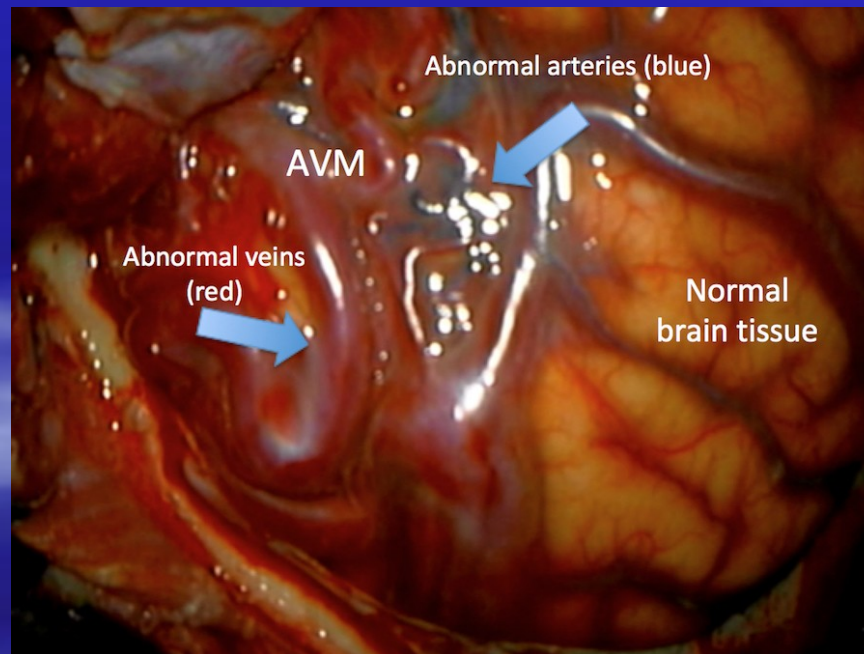
Clinical presentation

- Hemorrhage
- Seizure
- Mass effect
- Ischemia : by steal
- Headache



Severity of disease

- Clinical presentation as the glasgow coma scale score



management

- AVM would be treated when presented with serious problem such as hemorrhage.
- Cavernoma need to be treated when present with multiple bleeding
- Other types of AVM wouldn't need to be treated

Spetzler Martin grade

Table 1. Spetzler–Martin Grading Scale for Arteriovenous Malformations of the Brain.*

Lesion Characteristic	Points
Size	
Small (<3 cm)	1
Medium (3–6 cm)	2
Large (>6 cm)	3
Location	
Noneloquent	0
Eloquent	1
Veins	
Superficial	0
Deep	1

* The scale is used to make decisions about treatment risks while taking into account the anatomical characteristics of a cerebral arteriovenous malformation. The grade is the sum of points assigned to the size of the lesion, the location (in or not in the eloquent cortex), and the presence or absence of deep venous drainage.

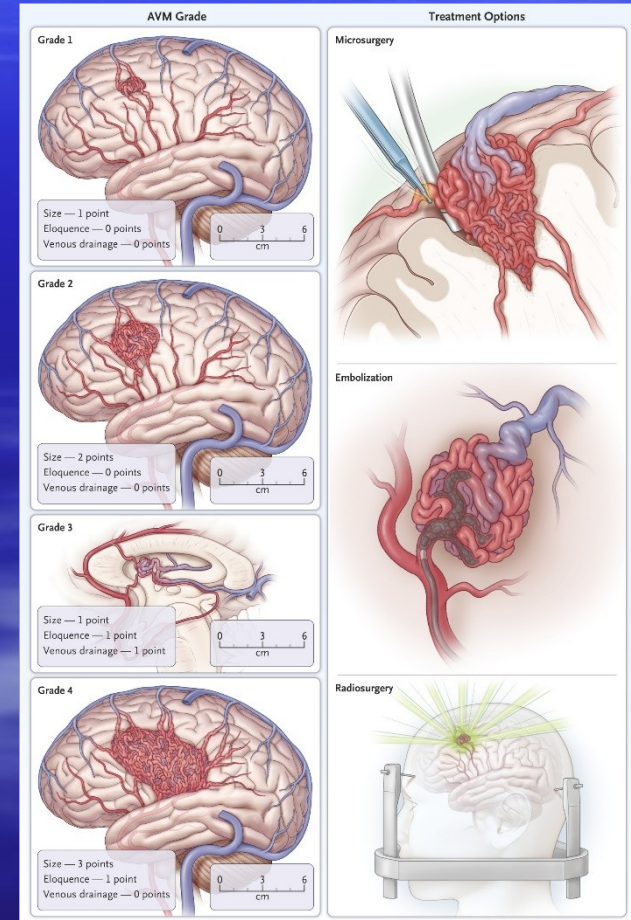
Table 2. Overall Favorable Outcomes of Surgical Removal and Radiosurgery for the Treatment of Cerebral Arteriovenous Malformation, According to the Spetzler–Martin Grade.*

Spetzler–Martin Grade	Favorable Outcome	
	Microsurgery	Radiosurgery
	<i>percent of patients</i>	
1	95	70
2	95	70
3	80	56
4	30	35
5	30	35

* Favorable outcome after microsurgery is defined as complete cure of the arteriovenous malformation, according to angiography, with no neurologic deficit that would prevent the patient from resuming all normal activities and full-time employment.¹⁴⁻¹⁶ Favorable outcome after radiosurgery is defined as obliteration of the arteriovenous malformation (confirmed by MRI or angiography), without post-treatment hemorrhage or permanent symptomatic radiation-induced brain injury.¹¹

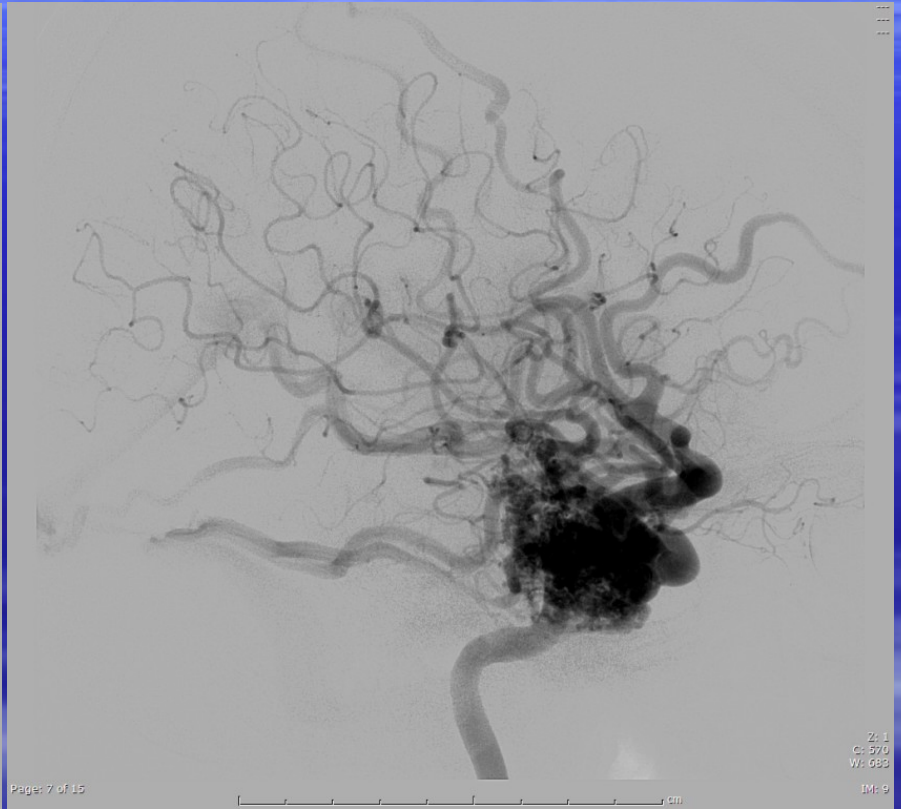
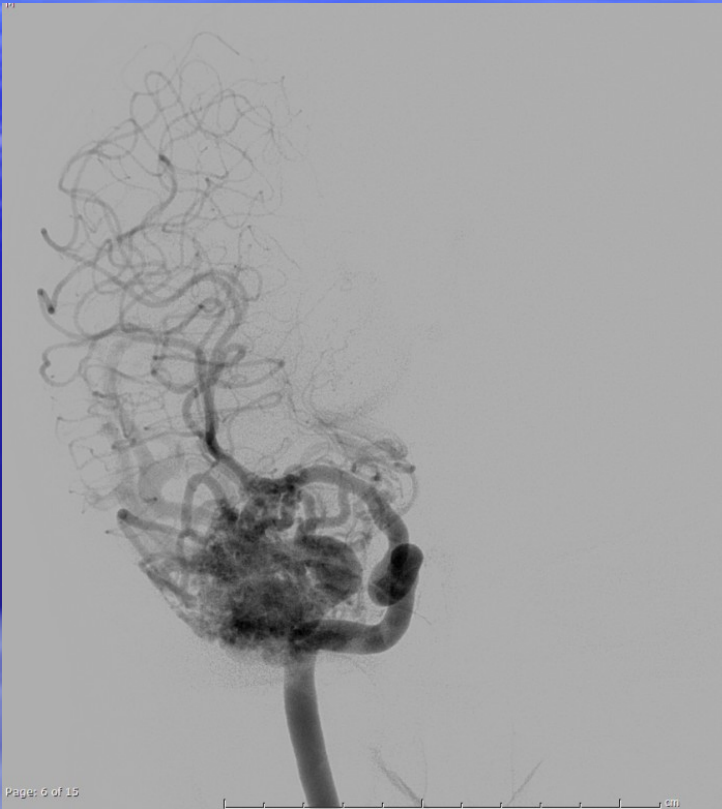
Choice of treatment

- Surgery
- Endovascular surgery
- Radiosurgery



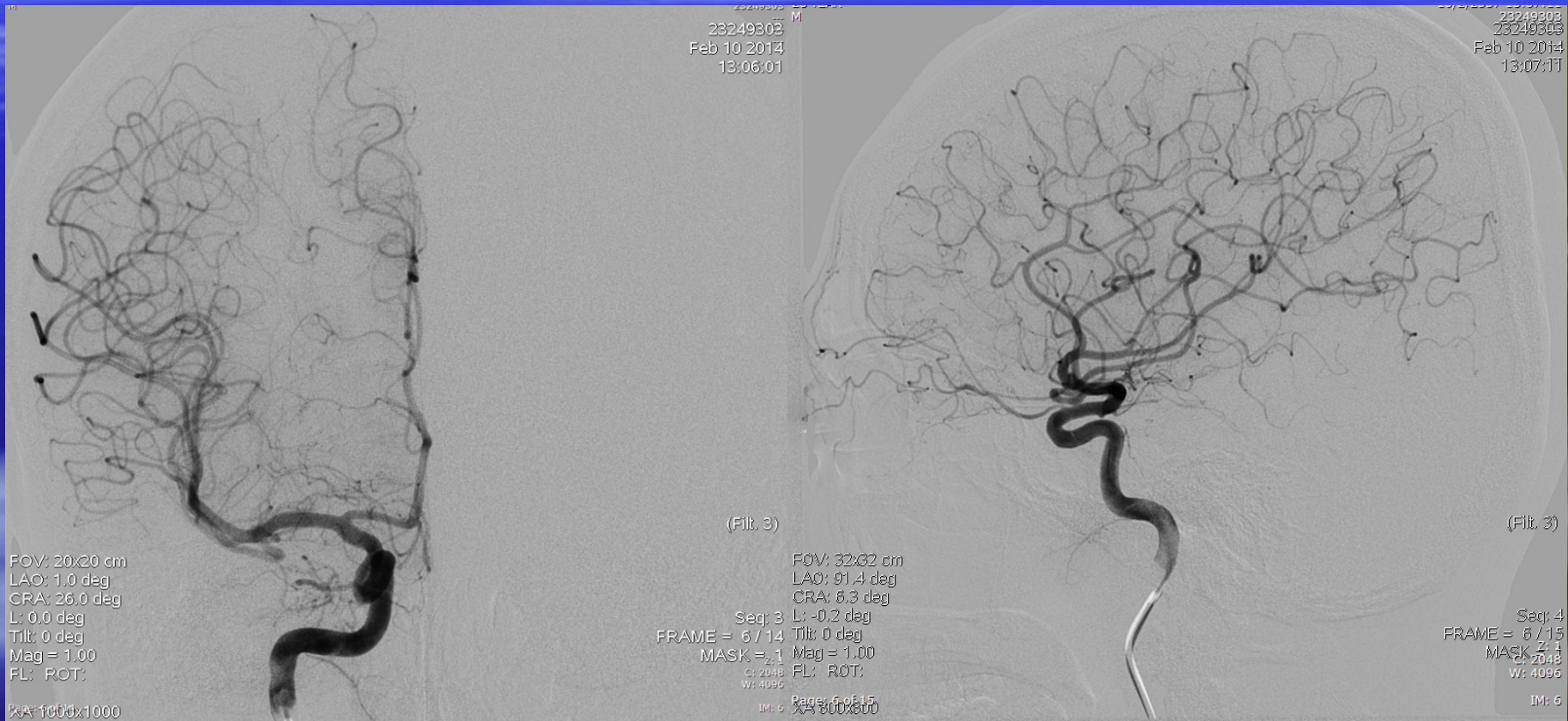


Before Surgery





After surgery



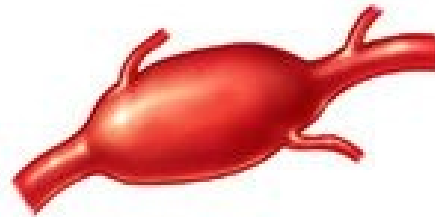
Postoperation

- Rebleeding
- Seizure
- Perfusion breakthrough bleeding

Cerebral aneurysm



Saccular Aneurysm



Fusiform Aneurysm



Ruptured Aneurysm

Clinical presentation

- Mass effect
- Rupture present Subarachnoid hemorrhage or intracerebral hematoma
- Emboli phenomenon

Subarachnoid hemorrhage

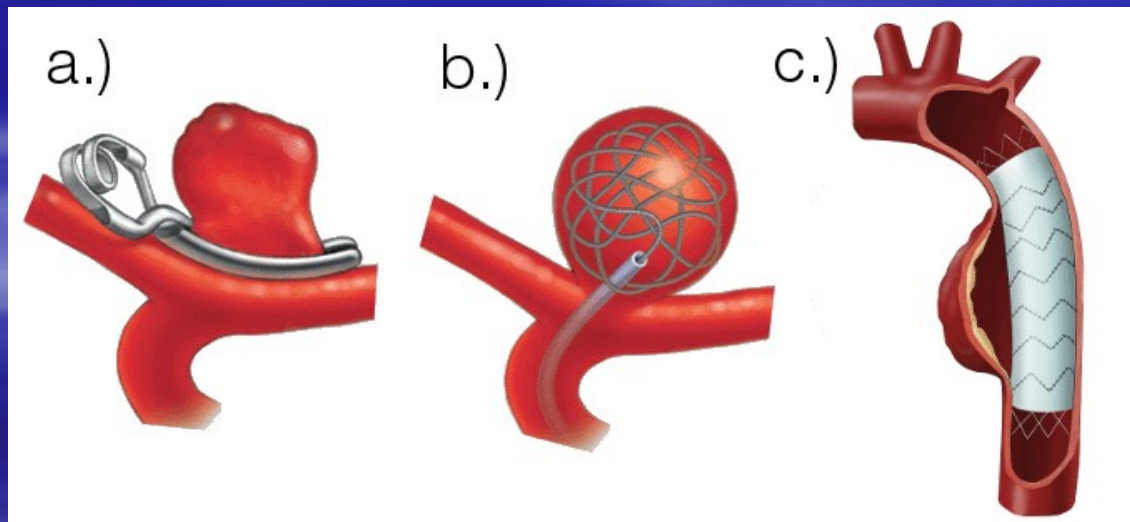
- Common caused by ruptured cerebral aneurysm
- Severity depends on Glasgow coma scale score
- Mortality could be up to 30%

SAH effect

- Rebleeding
- Intracranial hypertension
- Hydrocephalus
- Vasospasm
- Seizure
- Electrolyte imbalance\
- Arrhythmia

Option of treatment

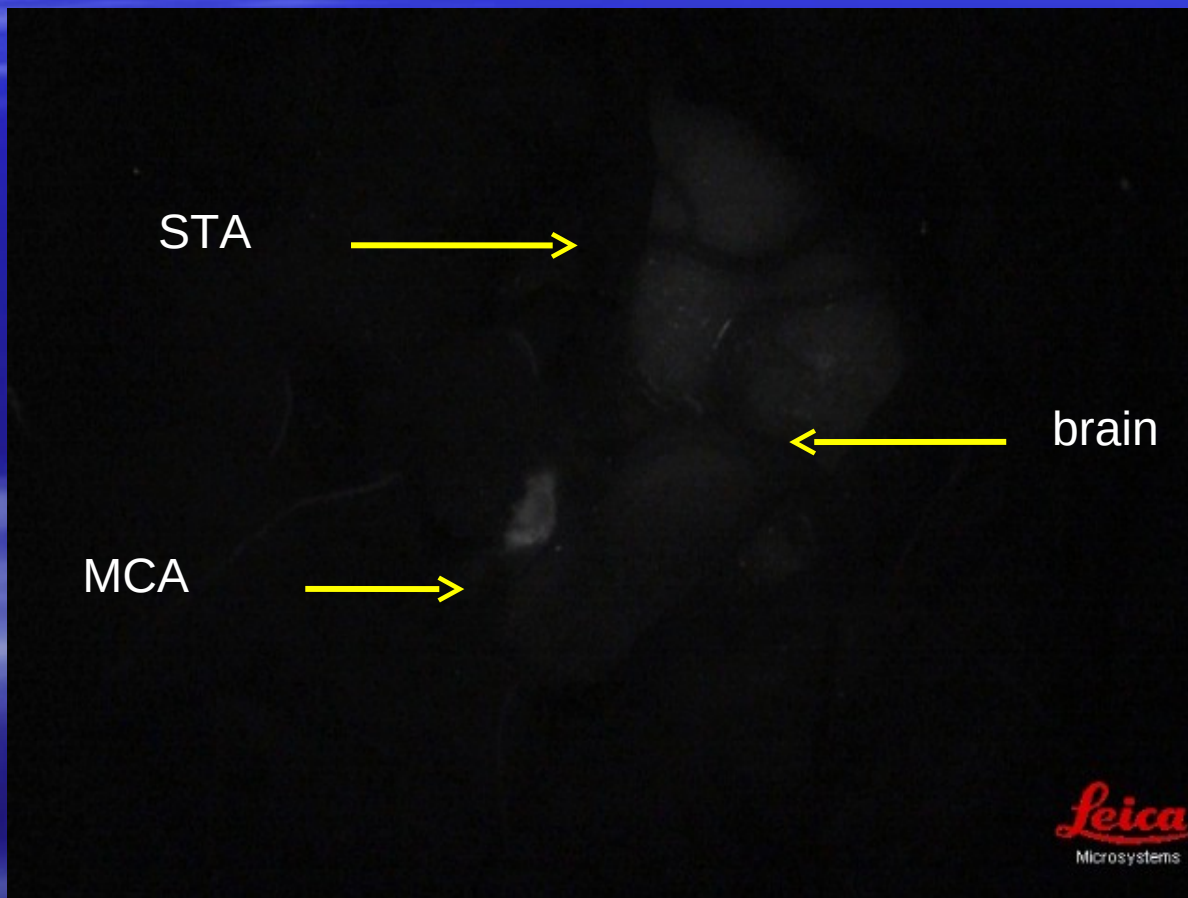
- Surgery: clipping, trapping, wrapping, bypass
- Endovascular: coiling, stenting, coiling with stenting





Intraoperative imaging

- ICG(indocyanine green) injection





Hybrid Theater





Hybrid theater



Vasospasm treatment

- Common occur at day 3 rd and last for 3 weeks
- 3 H therapy: hemodilution, hypertension, hypervolemia
- Ca^{2+} channel blocker: Nimodipine(nimotop)
- Transcranial doppler monitoring

Prognosis of SAH

- Depend on clinical presentation
- Location of cerebral aneurysm

Thank you for your attention